

Modelling Crypto Asset Order-Flow Imbalance as an Additive and Multiplicative Process

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Abstract

This study models order-flow imbalance (OFI) in cryptocurrency markets as a stationary additive multiplicative stochastic process and derives a structural decomposition of conditional return variance into an additive and multiplicative OFI channel. The multiplicative component captures the self-amplifying dynamics generated by trend followers, leveraged traders, and large holders, whose state-dependent order flow pressure defines concentrated market regimes.

Estimated on five major cryptocurrency pairs over 2019–2024, the multiplicative variance component $\hat{\sigma}_m^2$ is positive and significant across twelve independent six-month sub-periods for all five assets (Fama–MacBeth t -statistics 3.26–5.05), establishing state-dependent OFI amplification as a persistent structural feature of the cryptocurrency order flow. The multiplicative share $\hat{s}_t^{(m)}$ leads realised volatility by up to eighty hours around crisis events built on concentrated directional pressure, including the COVID crash and the LUNA collapse, but does not generate an unconditional timing signal — a result formalised via a Kolmogorov–Smirnov test against the geometric null. Transmission of multiplicative OFI dynamics into return variance is gated by market depth: during the FTX collapse, when thin depth and an elevated multiplicative regime coincided, the OFI channel explained 10.6 percent of realized squared returns; in all other episodes, it explained less than one percent. The dominance threshold $\hat{\sigma}_a/\hat{\sigma}_m$, ranging from 0.227 (XRP) to 0.430 (ETH), varies systematically with market depth and ownership concentration, providing a continuous model-derived measure of when market dynamics are driven by state-dependent amplification rather than by background noise.

Keywords: Order-flow imbalance, Additive multiplicative process, Price impact, Market depth, Cryptocurrency markets, Heavy tails, Multiplicative noise, Regime detection

1 Introduction

Heavy tails in asset return distributions are a well-established empirical regularity (Mandelbrot, 1963; Fama, 1965; Cont, 2001). This study shifts the focus away from returns and studies the order-flow imbalance of crypto assets (Cont et al., 2014) as a stochastic object of independent interest. This study was motivated by three factors. First, while order flow imbalance has been widely used as an explanatory variable in market microstructure (Anastasopoulos et al., 2026; Xu et al., 2019; Jaisson, 2015), this study is the first to model it as a stationary additive multiplicative process. Second, this specification provides a direct structural link between order flow dynamics, market depth, and the fat tails of the return distribution: heavy-tailed returns emerge as a consequence of the order flow mechanism rather than as a distributional assumption. Third, the multiplicative component yields a time-varying estimate of the dominance of large holders (“whales”), whose state-dependent order-flow pressure defines the multiplicative regime. These features make the framework particularly suited to crypto markets, where institutional participation, concentrated on-chain wealth, and episodic liquidity stress make order flow dynamics central to price formation.

Our approach is related to the literature on order flow toxicity and the volume-synchronized probability of informed trading (VPIN), in which volume imbalance is used to construct a real-time measure of toxic order flow faced by liquidity providers (Easley et al., 2012). We build on the same microstructural intuition but pursue a different objective and operate at lower frequencies. Rather than using imbalance as an indicator of adverse selection or impending liquidity stress, we model the imbalance process itself as a stochastic object whose dynamics alternate between the additive and multiplicative regimes. More broadly, the linear price-impact assumption that underlies optimal execution models, such as Almgren and Chriss (2001), provides a useful benchmark. However, it is precisely in the multiplicative regime, when concentrated, self-reinforcing order flow dominates, that this linearity breaks down most severely, and the present framework becomes most relevant.

We model the OFI as an additive multiplicative stochastic process (Takayasu et al., 1997; Nakao, 1998; Sato, 2004; Lux, 2006; Sato, 2010). Statistically, this specification generates heavy-tailed distributions naturally, a feature that characterises order-flow variables no less than returns. Economically (Yakovenko, 2016), it captures the interaction between *additive noise traders*, whose impact is state-independent, and *multiplicative noise traders*, whose impact scales with the prevailing OFI level. The latter amplify one-sided order flows and correspond to three well-documented participant types: trend followers, who reinforce the prevailing price direction (De Long et al., 1990; Cutler et al., 1990); leveraged traders, who unwind into falling markets and generate self-reinforcing sell-side pressure (Geanakoplos, 2010; Brunnermeier and Pedersen, 2008); and large holders (whales) whose, position sizes produce order-flow pressure disproportionate to routine trading noise (Bouchaud et al., 2009; Makarov and Schoar, 2020; Ante, 2023). Therefore, the multiplicative component is a continuous, time-varying measure of the degree to which market dynamics are

driven by state-dependent amplification rather than background noise.

Following [Yakovenko \(2016\)](#), additive interactions produce thin-tailed equilibrium distributions, whereas multiplicative interactions generate power-law tails. Therefore, the emergence of fat tails in OFI is a structural consequence of the concentration of trading power rather than a purely behavioral phenomenon, and estimating the model parameters provides a continuous, real-time signal of when multiplicative noise traders dominate market dynamics.

The empirical analysis is conducted on five major cryptocurrency pairs — BTCUSDT, ETHUSDT, BNBUSDT, SOLUSDT, and XRPUSDT — using four-hour intraday bars from Binance over the period 2019–2024. The paper’s central structural result is the decomposition of the conditional return variance into an additive and multiplicative OFI channel. From the price-impact equation

$$r_{t+1} = \alpha + \beta I_t + \varepsilon_{t+1},$$

(For notational clarity, the introduction omits the UTC-hour subscript h ; the full notation $r_{i,t,h}$, $\beta_{i,h}$, $\varepsilon_{i,t,h}$ is used throughout Section 2.) the conditional variance of next-bar returns is

$$\text{Var}(r_{t+1} | I_t) \approx \beta_t^2 (\sigma_a^2 + \sigma_m^2 I_t^2) + \sigma_{\varepsilon,i}^2.$$

This expression, derived formally as equations (25) and (26), reveals that return variance directly inherits the additive multiplicative structure of the OFI process. When the current imbalance state I_t is large, the multiplicative term $\beta^2 \sigma_m^2 I_t^2$ dominates, and the return variance scales with the square of the prevailing OFI level. When I_t is small, the additive term $\beta^2 \sigma_a^2$ prevails, and the return variance is approximately constant. The fraction of forecast return variance attributable to the multiplicative OFI channel is

$$s_t^{(m,r)} = \frac{\beta^2 \sigma_m^2 I_t^2}{\beta^2 (\sigma_a^2 + \sigma_m^2 I_t^2) + \sigma_{\varepsilon}^2}.$$

This is equation (28). Large return realisations therefore require two forces to act simultaneously: an elevated OFI state that activates the multiplicative channel, and a large price-impact coefficient β that transmits that channel into prices — a condition that is equivalent to thin market depth by the argument of [Cont et al. \(2014\)](#).

The main empirical findings are as follows. First, the multiplicative variance component $\hat{\sigma}_m^2$ is positive and statistically significant for all five assets across 12 independent six-month sub-periods, establishing that state-dependent OFI amplification is a persistent structural feature of cryptocurrency order flow rather than an artifact of any single market episode. Second, the multiplicative share $\hat{s}_t^{(m)}$ activates episodically and leads to realized volatility by up to 80 hours

around major market events, including the COVID crash of March 2020 and the LUNA collapse of May 2022, providing evidence that the model detects structural fragility before it manifests in prices. Third, the transmission of multiplicative OFI dynamics into return variance is gated by market depth: the price-impact coefficient $\hat{\beta}_t$ collapses to near zero in deep markets, quarantining the multiplicative channel from prices, while in thin markets, the two forces act simultaneously, and the OFI channel explains a disproportionate share of realized return variance — ten percent during the FTX collapse, against less than one percent in all other episodes. Fourth, the forward-distance distribution from multiplicative regime onsets to large price moves is consistent with a memoryless geometric process, establishing that the model’s leading-indicator properties are episode-specific rather than unconditional and bounding the paper’s forecasting claims accordingly.

The remainder of this paper is organized as follows. Section 2 develops the additive multiplicative model of OFI, derives the conditional variance decomposition, and establishes a formal link between OFI dynamics and return variance through the price-impact equation. Section 3 presents the empirical results across six subsections: validation of the additive multiplicative structure via Fama-MacBeth estimation, episode analysis of regime activation, distributional consequences of the multiplicative regime, price-impact link and depth-absorption mechanism, timing analysis of large price moves, and cross-sectional heterogeneity across the five assets. Section 4 concludes.

2 Additive–Multiplicative Processes and Order-Flow Imbalance

2.1 Additive–Multiplicative Processes

We model the state variable of interest, denoted by x_t , as an additive-multiplicative stochastic process. In continuous time, a convenient specification is

$$dx_t = \nu x_t dt + \sigma_m x_t dW_t^{(m)} + \sigma_a dW_t^{(a)}, \quad (1)$$

where ν is the drift parameter, $\sigma_m \geq 0$ controls the multiplicative component, $\sigma_a \geq 0$ controls the additive component, and $W_t^{(m)}$ and $W_t^{(a)}$ are independent Wiener processes. The process combines a state-dependent random term, $\sigma_m x_t dW_t^{(m)}$, with a state-independent random term, $\sigma_a dW_t^{(a)}$.

Equation (1) is attractive because it captures two distinct mechanisms: the *additive* component represents background shocks whose magnitude does not depend on the current state. In contrast, the *multiplicative* component scales with x_t itself, so that large realizations of the state generate proportionally larger future fluctuations. In the context of order flow imbalance, this means that one-sided flow can become self-reinforcing when the multiplicative channel is strong.

Why additive–multiplicative dynamics generate fat tails. A simple discrete-time approximation to (1) is

$$x_{t+1} = x_t + \nu x_t \Delta + \sigma_m x_t \sqrt{\Delta} \varepsilon_t + \sigma_a \sqrt{\Delta} \zeta_t, \quad (2)$$

where ε_t and ζ_t are independent, standard normal shocks. Rearranging,

$$x_{t+1} = a_t x_t + b_t, \quad a_t := 1 + \nu \Delta + \sigma_m \sqrt{\Delta} \varepsilon_t, \quad b_t := \sigma_a \sqrt{\Delta} \zeta_t. \quad (3)$$

This is a random recurrence equation of the Kesten type. Its key feature is that the next state is obtained by multiplying the current state by a random coefficient a_t and then adding an innovation b_t . Under broad conditions, such processes exhibit stationary distributions with power-law tails rather than Gaussian tails. Intuitively, the additive term repopulates the center of the distribution, whereas the multiplicative term occasionally amplifies large states and generates rare but extreme realizations. It is precisely this interaction between state-independent replenishment and state-dependent amplification that gives rise to fat-tailed behavior (Kesten, 1973; Takayasu et al., 1997; Nakao, 1998; Sato, 2010).

In a continuous-time setting, this mechanism appears directly in the conditional variance. From (1), the local conditional mean and variance satisfy

$$\mathbb{E}[\Delta x_t \mid x_t] \approx \nu x_t \Delta, \quad (4)$$

and

$$\text{Var}(\Delta x_t \mid x_t) \approx \Delta (\sigma_a^2 + \sigma_m^2 x_t^2). \quad (5)$$

Equation (5) makes the fat-tail mechanism transparent. When $|x_t|$ is small, the additive component dominates, and the process behaves like a conventional noise-driven system. When $|x_t|$ is large, the multiplicative term $\sigma_m^2 x_t^2$ increases the conditional variance, making extreme states more likely to be followed by further large moves in the stock price.

Estimating additive and multiplicative components. Given a time series $\{x_t\}_{t=1}^T$, a simple two-step estimator follows from (4)–(5). Define first differences as

$$\Delta x_t := x_{t+1} - x_t \quad (6)$$

Then estimate the conditional mean equation

$$\Delta x_t = \alpha + \nu x_t + u_t, \quad (7)$$

where u_t is the residual. In a second step, estimate the variance equation

$$u_t^2 = \sigma_a^2 + \sigma_m^2 x_t^2 + \varepsilon_t. \quad (8)$$

The intercept in (8) provides an estimate of the additive variance component, whereas the slope on x_t^2 provides an estimate of the multiplicative variance component. In practice, because variance parameters must be nonnegative, one may impose $\sigma_a^2 \geq 0$ and $\sigma_m^2 \geq 0$ directly, or clip negative estimates at zero, as follows:

Share of the multiplicative component. A useful summary statistic is the fraction of the conditional variance attributable to the multiplicative channel at state x_t . From (5), the multiplicative share is

$$s_t^{(m)} := \frac{\sigma_m^2 x_t^2}{\sigma_a^2 + \sigma_m^2 x_t^2}, \quad (9)$$

while the additive share is

$$s_t^{(a)} := \frac{\sigma_a^2}{\sigma_a^2 + \sigma_m^2 x_t^2}. \quad (10)$$

By construction, $s_t^{(m)} + s_t^{(a)} = 1$. These quantities have natural interpretations. When $s_t^{(m)}$ is close to zero, the conditional variance is driven primarily by additive shocks. When $s_t^{(m)}$ is large, fluctuations are increasingly state-dependent, and the system is in a multiplicative regime. Using estimated parameters, the empirical multiplicative share is

$$\hat{s}_t^{(m)} = \frac{\hat{\sigma}_m^2 x_t^2}{\hat{\sigma}_a^2 + \hat{\sigma}_m^2 x_t^2}. \quad (11)$$

A smoothed version, such as a moving average of $\hat{s}_t^{(m)}$, can be interpreted as a regime indicator for the relative importance of state-dependent amplification, over time.

Threshold for multiplicative dominance. A simple threshold separates additive-dominated and multiplicative-dominated states. Setting the two variance contributions equal yields $\sigma_m^2 x_t^2 = \sigma_a^2$, so the critical magnitude is

$$|x_t| = \frac{\sigma_a}{\sigma_m}. \quad (12)$$

For $|x_t| < \sigma_a/\sigma_m$, the additive noise dominates. For $|x_t| > \sigma_a/\sigma_m$, the multiplicative component contributes more to the conditional variance than the additive component. This threshold provides a simple criterion for characterizing when the system enters the self-amplifying regime.]

2.2 Order-Flow Imbalance

Order-flow imbalance (henceforth, OFI) is a reduced-form measure of the directional pressure exerted by buyers (sellers) over a given interval. In the market microstructure, OFI is closely related to the idea that prices respond not only to the arrival of information but also to the imbalance between aggressive buying (selling) pressure (Cont et al., 2014; Jaisson, 2015; Xu et al., 2019; Easley et al., 2012). A positive imbalance indicates that the buy-initiated flow dominates the sell-initiated flow and vice versa. In this sense, OFI summarizes the directional composition of trading activity and is a natural state variable for the short-horizon price formation.

Construction from crypto market data. A useful feature of crypto markets is that exchange data¹ often make it possible to construct OFI directly from high-frequency trading records or fine time-bar data. In our setting, the raw data consist of open, high, low, close, and volume (OHLCV) observations, together with quote-volume and taker-buy volume.

Let QV_t denote the quote volume over interval t , and let $TBQV_t$ denote the taker-buy quote volume over the same interval. Then taker-sell quote volume is given by

$$TSQV_t = QV_t - TBQV_t \quad (13)$$

A natural raw imbalance measure is therefore

$$OFI_t^{\text{raw}} = TBQV_t - TSQV_t = 2 TBQV_t - QV_t \quad (14)$$

Because the absolute magnitude of (14) scales with market activity, it is often useful to normalize it by the total quote volume. This yields the relative order-flow imbalance

$$I_t = \frac{OFI_t^{\text{raw}}}{QV_t} = \frac{TBQV_t - TSQV_t}{QV_t} = \frac{2 TBQV_t - QV_t}{QV_t}, \quad (15)$$

which lies in the interval $[-1, 1]$. Values near 1 correspond to intervals dominated by aggressive buying, values near -1 correspond to intervals dominated by aggressive selling, and values near 0 indicate a relatively balanced flow.

Relation to market microstructure. The OFI is naturally connected to the core mechanisms of the market microstructure. Prices are formed through the interaction of aggressive orders, passive liquidity, and inventory adjustments by liquidity providers. When aggressive buy volume persistently exceeds aggressive sell volume, liquidity at the ask is depleted, and prices tend to move upward; the reverse occurs when sell pressure dominates. Therefore, OFI captures a key channel

¹In this paper – Binance

through which order flow affects short-run price changes (Cont et al., 2014). It is also closely related to the notion of order flow toxicity, in which a persistent imbalance exposes liquidity providers to adverse selection risk (Easley et al., 2012). In this sense, OFI is not merely a statistical transformation of trading volume but a microstructural variable that summarizes the directional pressure faced by the limit order book.

Why OFI is a natural state variable. The central premise of this study is that OFI can be treated not only as an explanatory variable for returns but also as a stochastic process in its own right. This is especially relevant in crypto markets, where fragmented liquidity, leverage, large traders, and liquidation cascades can generate a persistent one-sided flow. If routine trading produces additive shocks, while whales, forced liquidations, or trend-following behavior amplify existing imbalances multiplicatively, then the OFI itself may become heavy-tailed and regime-dependent. Therefore, directly modelling the OFI links microstructure-based order flow analysis with stochastic process models of fat-tailed market dynamics.

2.3 A Formal Model of the OFI–Return Relationship

The additive multiplicative process describes the evolution of the order flow imbalance state variable, whereas the contemporaneous price-impact regression links this state variable to returns. We model the dynamics of the relative OFI (I_t) as an additive multiplicative process by restating (1) as follows:

$$dI_t = \nu I_t dt + \sigma_m I_t dW_t^{(m)} + \sigma_a dW_t^{(a)}, \quad (16)$$

And, similarly to (2)

$$\Delta I_t = \nu I_t \Delta + \sigma_m I_t \sqrt{\Delta} \varepsilon_t^{(m)} + \sigma_a \sqrt{\Delta} \varepsilon_t^{(a)}, \quad (17)$$

Hence,

$$\mathbb{E}[\Delta I_t | I_t] = \nu I_t \Delta, \quad (18)$$

$$\text{Var}(\Delta I_t | I_t) = \Delta (\sigma_a^2 + \sigma_m^2 I_t^2). \quad (19)$$

Equation (19) is the key implication of the additive-multiplicative model. When $|I_t|$ is small, the conditional variance of the OFI changes is dominated by the additive component σ_a^2 . When $|I_t|$ is large, the multiplicative component $\sigma_m^2 I_t^2$ becomes dominant, implying that extremely imbalanced states are associated with larger subsequent fluctuations in the imbalance.

Price-impact equation. To connect the imbalance state to returns, we use a contemporaneous price-impact equation:

$$r_t = \alpha + \beta I_t + \varepsilon_t, \quad (20)$$

where r_t denotes log-return, β measures return sensitivity to order-flow imbalance.

Equation (16) governs how the relative imbalance evolves, while Equation (20) maps the imbalance into contemporaneous returns. Combining the two equations provides a channel through which additive multiplicative OFI dynamics can affect returns.

Conditional return moments. From (20), the expected log-return given current relative imbalance is

$$\mathbb{E}[r_t \mid I_t] = \alpha + \beta I_t, \quad (21)$$

provided that $\mathbb{E}[\varepsilon_t \mid I_t] = 0$. Thus, the (relative) OFI-return beta determines the average contemporaneous return response to a given imbalance.

The unconditional variance of returns is

$$\begin{aligned} \text{Var}(r_t) &= \text{Var}(\alpha + \beta I_t + \varepsilon_t) \\ &= \beta^2 \text{Var}(I_t) + \text{Var}(\varepsilon_t) + 2\beta \text{Cov}(I_t, \varepsilon_t). \end{aligned} \quad (22)$$

If the return innovation is orthogonal to OFI, so that $\text{Cov}(I_t, \varepsilon_t) = 0$, then

$$\text{Var}(r_t) = \beta^2 \text{Var}(I_t) + \sigma_\varepsilon^2. \quad (23)$$

Equation (23) shows that any increase in the variance or tail thickness of I_t is transmitted to the returns in proportion to β^2 . Hence, if the additive multiplicative dynamics generate heavy-tailed OFI, the price-impact equation provides a direct mechanism through which these OFI tails can contribute to heavy-tailed returns.

Conditional return variance. The price-impact Equation (20) maps the next-period imbalance state into next-period returns. Applying it one period forward,

$$r_{t+1} = \alpha + \beta I_{t+1} + \varepsilon_{t+1}. \quad (24)$$

Taking the variance of (24) conditional on the current imbalance state I_t , and using the orthogonality assumption $\mathbb{E}[\varepsilon_{t+1} \mid I_t] = 0$, gives

$$\text{Var}(r_{t+1} \mid I_t) = \beta^2 \text{Var}(I_{t+1} \mid I_t) + \sigma_\varepsilon^2. \quad (25)$$

The term $\text{Var}(I_{t+1} \mid I_t)$ is the conditional variance of the next OFI state given the current one. From the additive multiplicative dynamics (19), this equals $\Delta(\sigma_a^2 + \sigma_m^2 I_t^2)$. Substituting,

$$\text{Var}(r_{t+1} \mid I_t) \approx \beta^2 \Delta(\sigma_a^2 + \sigma_m^2 I_t^2) + \sigma_\varepsilon^2. \quad (26)$$

Equation (26) provides a formal link between the additive multiplicative OFI dynamics and state-dependent return variation. The current imbalance state I_t forecasts the variance of the *next* bar's return through two channels: the additive component σ_a^2 , which captures the baseline return variance inherited from routine imbalance fluctuations, and the multiplicative component $\sigma_m^2 I_t^2$, which amplifies the variance when the current imbalance state is large. The price-impact coefficient β scales both channels equally, such that a high-beta asset translates OFI dynamics into return variance more strongly than a low-beta asset.

Multiplicative share of return variance. The multiplicative share of OFI variance at state I_t is

$$s_t^{(m)} = \frac{\sigma_m^2 I_t^2}{\sigma_a^2 + \sigma_m^2 I_t^2}. \quad (27)$$

Combining this with (26), the fraction of $\text{Var}(r_{t+1} \mid I_t)$ attributable to the multiplicative OFI channel is

$$s_t^{(m,r)} = \frac{\beta^2 \Delta \sigma_m^2 I_t^2}{\beta^2 \Delta(\sigma_a^2 + \sigma_m^2 I_t^2) + \sigma_\varepsilon^2}. \quad (28)$$

This is the share of the next-bar return variance that the current imbalance state I_t contributes through the multiplicative channel. It is amplified by three forces: a large multiplicative OFI parameter σ_m , a large current imbalance state $|I_t|$, and a large price impact coefficient β . Conversely, when σ_m is small, $|I_t|$ is close to zero, or β is small, the multiplicative OFI channel contributes less to the forecast return variance.

The role of market depth. The price-impact coefficient β is inversely related to market depth (Cont et al., 2014), so an elevated multiplicative OFI, $s_t^{(m)}$, need not translate into a large return-variance share, $s_t^{(m,r)}$, when liquidity is sufficient to absorb the flow. Large returns require both concentrated, self-amplifying order flows and thin depths, reflected in a large $|\hat{\beta}|$. Thus, with ample depth, the multiplicative OFI channel remains largely insulated from prices; with thin depth, the same flow moves prices sharply, and vice versa. This depth-conditioning helps explain the muted $\hat{s}_t^{(m,r)}$ during

the 2024 institutional bull run despite elevated $\hat{s}_t^{(m)}$, as well as the negative predictive coefficient $\hat{b}_i$ in Table 8, consistent with episodes in which concentrated flow is absorbed and then quickly reverts.

Threshold for multiplicative dominance. The threshold for multiplicative dominance in the OFI process is obtained by setting the additive and multiplicative variance components equal: $\sigma_a^2 = \sigma_m^2 I_t^2$. Thus,

$$|I_t| = \frac{\sigma_a}{\sigma_m}. \quad (29)$$

For imbalance states satisfying $|I_t| > \sigma_a/\sigma_m$, the multiplicative component dominates the OFI variance. Through the price-impact equation, these are precisely the states in which imbalance shocks are most likely to generate large return movements, especially when β is large.

Summary. In summary, the additive multiplicative model and price-impact regression describe two linked parts of the same mechanism. The additive multiplicative process governs the evolution of I_t and determines how the current imbalance state forecasts the variance of the *next* bar’s return via (26). The price-impact coefficient β scales both the additive and multiplicative channels equally, transmitting OFI dynamics into the return variance. Large return realizations at $t + 1$ are therefore most likely when the current OFI state at t places the system in a multiplicative regime and the corresponding beta is high.

3 Empirical Results

This section presents the empirical results in six steps: First, we validate the additive–multiplicative structure of the OFI and show that both variance components are statistically significant across sub-periods. We then examine when the multiplicative regime activates, its episodic nature, and its link to major market events in the following section. Next, we study how regime activation shapes the return distribution before tracing its transmission into return variance through a time-varying price-impact coefficient. We then test whether multiplicative spikes precede large price changes. Finally, we examine cross-sectional differences across the five assets and link them to variations in liquidity, market structure, and ownership concentration.

Baseline OFI–Return Relationship

Figure 1 plots the pooled relationship between raw relative OFI (I_t , the signed volume-imbalance ratio defined in equation (15)) and raw four-hour log-returns in basis points for each of the five assets over the full sample 2019–2024. The relationship is positive and approximately linear in the central

region of the distribution, confirming the basic microstructure prediction that buy-side (sell-side) order flow pressure is associated with positive (negative) contemporaneous price movements.

In the extreme tails, however, the LOWESS smoother flattens or reverses: very large positive or negative imbalances do not always translate into proportionally larger price movements, indicating that the marginal price impact of OFI is state-dependent at large imbalance values. This tail behavior is consistent with the additive multiplicative model in Section 2.3, in which concentrated directional flow generates self-amplifying order-flow dynamics that a simple linear specification does not capture.

Table 1 reports the pooled OLS estimates of the contemporaneous return response to the raw OFI. The price-impact coefficient $\hat{\beta}$ is positive and highly significant for every asset ($t > 27$, $p < 0.001$ throughout), confirming a robust positive relationship between relative OFI and four-hour returns across all five markets. The coefficient ranges from 348.1 basis points per unit OFI for BNBUSDT to 1,255.8 basis points for SOLUSDT. Section 3.4 estimates the time-varying counterpart using a 250-bar rolling window, allowing the price impact coefficient to vary over the sample period in response to changing market conditions.

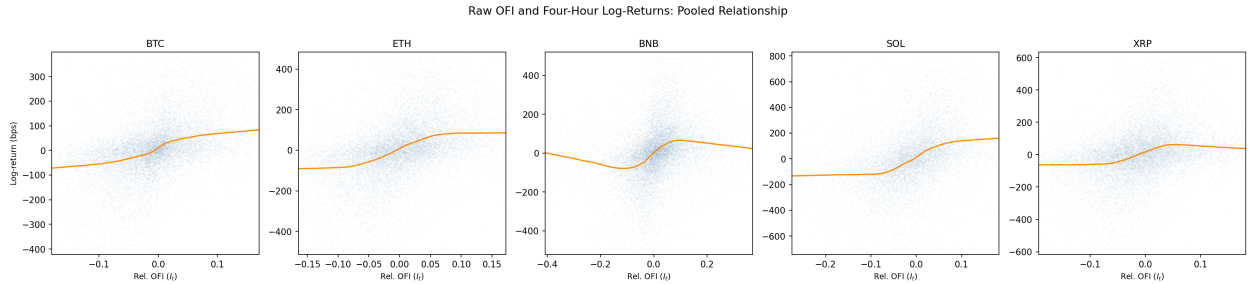


Figure 1: Raw relative OFI (I_t) and four-hour log-returns (basis points) for each of the five assets, pooled across the full sample 2019–2024. The scatter is plotted at $\alpha = 0.05$; both axes are clipped at the 1st and 99th percentiles for improved visual clarity. The dark orange curve is a LOWESS smoother (bandwidth = 0.3). The relationship is positive and approximately linear in the centre of the distribution; the flattening of the LOWESS curve in The tails are consistent with the state-dependent price impact.

3.1 Validating the Additive–Multiplicative Structure of OFI

To assess whether the additive multiplicative structure of the OFI is a persistent feature of the data rather than an artifact of a single market episode, we adopt the two-pass estimation procedure of Fama and MacBeth (1973). The sample (2019–2024) is divided into $K = 12$ equal-observation sub-periods of approximately six months. The two-step OLS estimator described in Section 2 is applied independently within each sub-period, yielding a time series of K parameter estimates

Table 1: Pooled OLS regression of raw four-hour log-returns on raw relative OFI, estimated over the full sample period of 2019–2024. The standard errors are HC3-robust. The coefficient $\hat{\beta}$ measures the average contemporaneous return response (in basis points) to a one-unit increase in relative OFI. Section 3.4 estimates the time-varying counterpart using a 250-bar rolling window.

Asset	$\hat{\beta}$	SE	t	p	R^2
BTC	810.828	20.141	40.26	< 0.001	0.1618
ETH	1,018.265	22.444	45.37	< 0.001	0.1600
BNB	348.054	12.838	27.11	< 0.001	0.0544
SOL	1,255.784	36.397	34.50	< 0.001	0.1477
XRP	776.546	21.182	36.66	< 0.001	0.0751

for each asset. The reported estimates are the cross-period means, and the reported t -statistics are Fama–MacBeth t -statistics,

$$t_{\text{FM}} = \frac{\bar{\theta}}{\hat{s}(\theta)/\sqrt{K}}, \quad (30)$$

where $\bar{\theta}$ is the cross-period mean, and $\hat{s}(\theta)$ is the cross-period standard deviation. A large $|t_{\text{FM}}|$ therefore reflects the consistency of the estimate across sub-periods, not merely precision over the full sample.

Results. Table 2 reports the Fama–MacBeth estimates for all five assets. Several findings are noteworthy.

First, $\hat{\nu}$ is negative and highly significant for every asset, with t -statistics ranging from -22.9 (BNB) to -60.7 (ETH). Mean reversion in OFI holds uniformly across all sub-periods and all assets, confirming the stationarity of the process in every segment of the sample.

Second, and most importantly, $\hat{\sigma}_m^2$ is positive and significant for every asset, with Fama–MacBeth t -statistics ranging from 3.26 (BTC) to 5.05 (XRP). The null hypothesis of a pure additive process ($\sigma_m^2 = 0$) is therefore rejected consistently over time across all five assets. This is the central empirical result of this study: multiplicative dynamics in OFI are not driven by a single market episode but constitute a persistent structural feature of cryptocurrency order flow across the entire 2019–2024 sample.

Third, the dominance threshold $\hat{\sigma}_a/\hat{\sigma}_m$ is well-identified and economically interpretable, ranging from 0.227 for XRP to 0.430 for ETH. The net directional OFI imbalance must exceed approximately 23–43 percent before the multiplicative channel contributes more to the conditional OFI variance than the additive channel. The lower threshold for XRP ($t = 11.87$) reflects the greater sensitivity of its order flow to concentrated directional pressure, consistent with the asset’s more concentrated

ownership structure and exposure to idiosyncratic regulatory events. The higher threshold for ETH ($t = 6.02$) reflects the greater depth of its order book, which requires a larger imbalance before self-amplifying dynamics are established.

Finally, the R^2 of the variance equation is significant for ETH, BNB, SOL, and XRP (t -statistics between 2.33 and 2.57), although only marginally so for BTC ($t = 1.95$). The relatively lower fit for BTC is consistent with a more complex variance structure that the linear x_t^2 specification does not fully capture, likely reflecting the greater heterogeneity of BTC’s market participant composition across the sample period.

Table 2: Fama–MacBeth staggered estimates of the additive multiplicative OFI parameters (Fama and MacBeth, 1973). The sample (2019–2024) is divided into $K = 12$ equal-observation sub-periods of approximately six months each. The two-step OLS estimator is applied within each sub-period; reported values are cross-period means with Fama–MacBeth t -statistics (equation (30)). $|t| > 2$ indicates that the parameter is consistent across sub-periods. The threshold is $\hat{\sigma}_a/\hat{\sigma}_m$; sub-periods in which $\hat{\sigma}_m = 0$ are excluded from the threshold mean and t -statistic.

	$\hat{v}$	$t(\hat{v})$	$\hat{\sigma}_a^2$	$t(\hat{\sigma}_a^2)$	$\hat{\sigma}_m^2$	$t(\hat{\sigma}_m^2)$	threshold	$t(\text{thr.})$	R^2	$t(R^2)$
BTC	−0.841	−54.50	0.00371	7.15	0.0818	3.26	0.328	3.38	0.0141	1.95
ETH	−0.879	−60.74	0.00424	10.82	0.0330	3.63	0.430	6.02	0.0024	2.57
BNB	−0.785	−22.92	0.01009	3.56	0.0997	4.14	0.277	9.27	0.0204	2.33
SOL	−0.854	−34.51	0.00616	4.72	0.0611	3.99	0.418	3.55	0.0066	2.47
XRP	−0.775	−23.45	0.00414	9.77	0.1014	5.05	0.227	11.87	0.0164	2.53

3.2 Episode Analysis: When Does the Multiplicative Regime Activate?

Having established that the multiplicative component of OFI variance is a persistent and statistically robust feature of cryptocurrency order flow, we turn to the question of *when* this component becomes economically dominant. The multiplicative share $\hat{s}_t^{(m)}$, defined in Equation (11), provides a continuous indicator of the relative contribution of the state-dependent channel to the OFI variance at each point in time. We use this indicator to characterize the episodes in which the multiplicative regime is activated and to assess whether these episodes anticipate subsequent price instability.

The episodic character of the multiplicative regime. Figure 2 plots the smoothed multiplicative share alongside the BTC price series over the full sample. The indicator is compressed near zero for the vast majority of the time, consistent with the theoretical prediction that the additive channel dominates when the order flow imbalance is modest. Exceedances of the 80th-percentile threshold are episodic and concentrated, and the series exhibits long stretches of near-zero activity punctuated by sharp, short-lived spikes. This pattern is consistent across all five assets, as shown in Figures 3–6,

although the level and frequency of threshold crossings vary with the market structure, as discussed in Section 3.7.

Four structural episodes. Table 3 and Figure 7 decompose the BTC sample into four episodes that differ sharply in the characteristics of the multiplicative regime. The first episode, spanning the second half of 2019 and early 2020, is characterized by sustained but moderate elevation: 29.6 percent of bars exceed the 80th-percentile threshold, and the peak value of 0.623 is the highest in the sample, yet the concurrent price move is only -22.0 percent with a maximum drawdown of -40.6 percent. This pattern is consistent with a period of structural thinness in the order book, in which routine imbalance fluctuations are sufficient to trigger state-dependent amplification, but no single large shock is dominant.

The second episode encompasses the COVID-19 market crash in March 2020. Here, the multiplicative share reaches a local peak of 0.162 on February 3, 2020, 37 days before the Black Thursday price collapse of approximately 50 percent. The concurrent correlation between $\hat{s}_t^{(m)}$ and the realized price range on the crash date itself is -0.122 , indicating that the indicator had already retreated by the time the crash occurred. This timing is not coincidental: the pre-crash period was characterized by concentrated directional flow as market participants repositioned in response to early COVID-19 news, producing self-amplifying OFI dynamics that resolved into a crash several weeks later.

The third episode covers the collapse of FTX in November 2022. The peak of $\hat{s}_t^{(m)} = 0.343$ occurs on January 10, 2023, two months *after* the exchange’s insolvency became public. This reframes the FTX episode as a buy-side multiplicative event: the multiplicative regime activated during the recovery rally, as concentrated buyers re-entered a thin market rather than during the collapse itself. The FTX collapse was a sell-side event driven by forced liquidations, the speed of which overwhelmed the OFI channel’s ability to build up a sustained multiplicative signal beforehand.

The fourth episode covers 2024, during which BTC reached a new all-time high of approximately \$73,000 in March. This episode is qualitatively distinct from the first three: 26.9 percent of bars exceed the threshold over the entire year, with a sustained elevated baseline rather than a single sharp spike. The concurrent price return is $+121.1$ percent with a maximum drawdown of only -8.2 percent. This pattern is consistent with the structural shift in market participation associated with the approval of spot Bitcoin ETFs in January 2024: large, institutionally driven order flows created a persistently multiplicative OFI environment without the acute stress characteristics of the crisis episodes.

Lead-lag analysis. The timing evidence from the four-episode analysis is formalized in Table 4, which reports the cross-correlation between $\hat{s}_t^{(m)}$ and the realized price range, measured as $(H_t - L_t)/C_t$, where H_t , L_t , and C_t denote the high, low, and close of the four-hour bar, respectively, over a ± 20 -bar window centered on each of the eight annotated events. A negative peak lag indicates

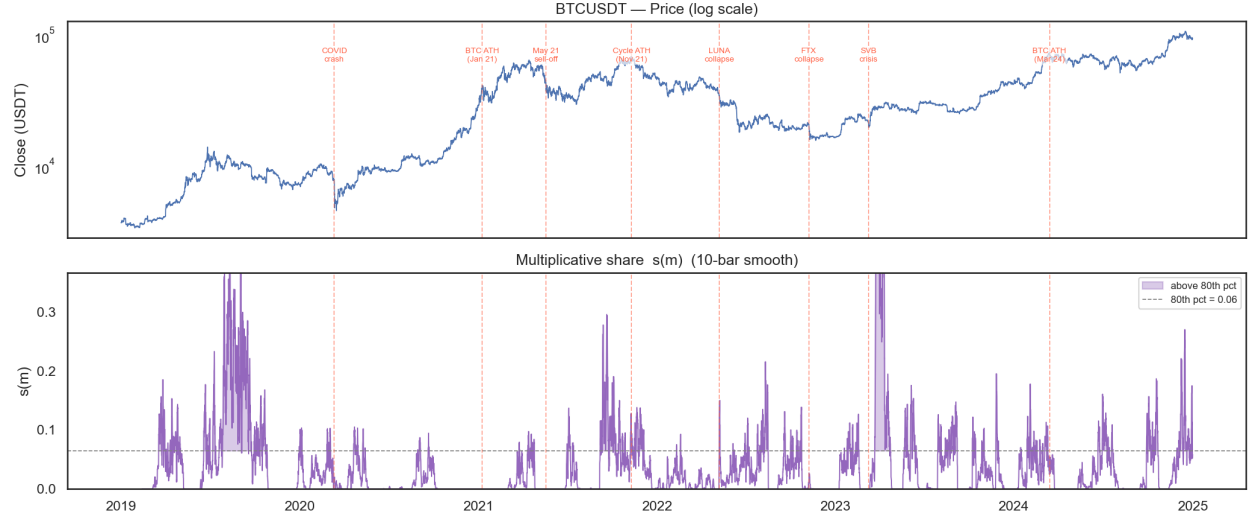


Figure 2: BTCUSDT price (log scale, top) and smoothed multiplicative share $\hat{s}_t^{(m)}$ (10-bar mean, bottom) over the full sample 2019–2024. The dashed line marks the 80th-percentile threshold ($\hat{s}_{0.80}^{(m)} = 0.07$). The shaded regions indicate the bars above the threshold. The vertical dashed lines mark the eight annotated market events.

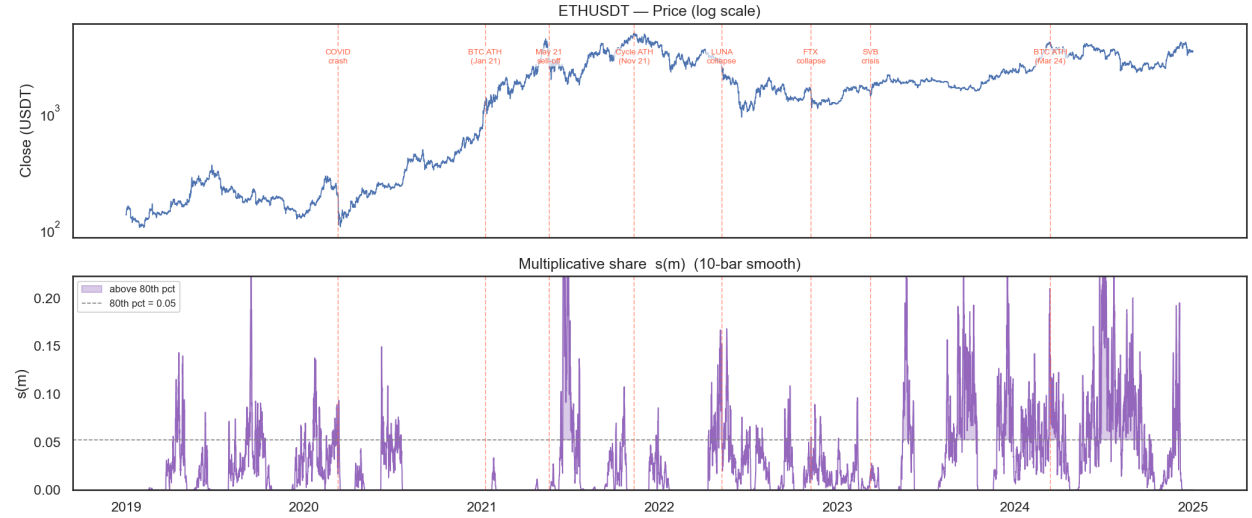


Figure 3: ETHUSDT price (log scale, top) and smoothed multiplicative share $\hat{s}_t^{(m)}$ (10-bar mean, bottom) over the full sample 2019–2024. The dashed line marks the 80th-percentile threshold ($\hat{s}_{0.80}^{(m)} = 0.05$).

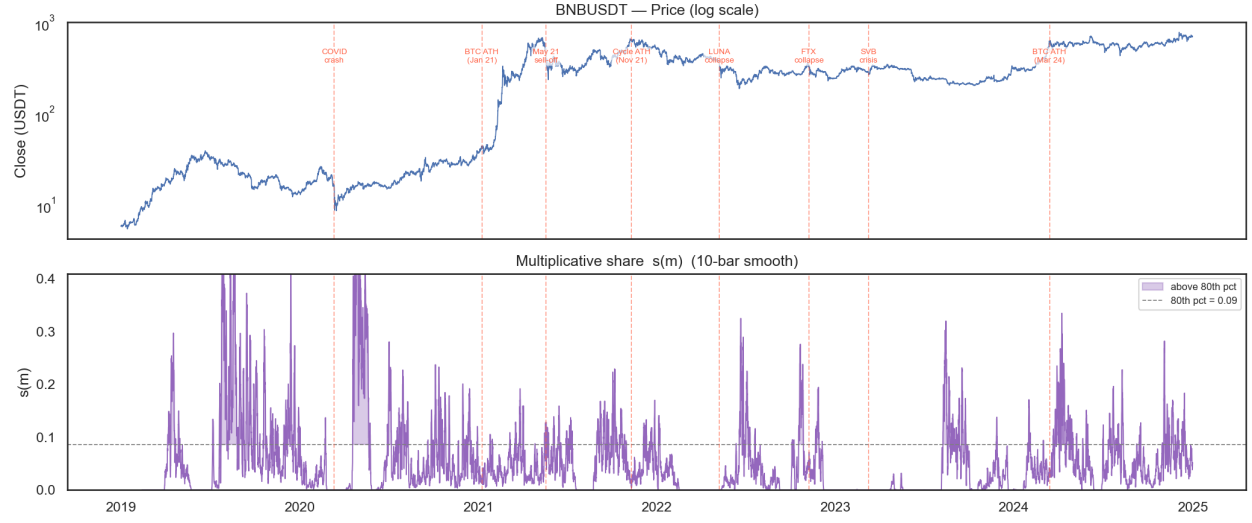


Figure 4: BNBUSDT price (log scale, top) and smoothed multiplicative share $\hat{s}_t^{(m)}$ (10-bar mean, bottom) over the full sample 2019–2024. The dashed line marks the 80th-percentile threshold ($\hat{s}_{0.80}^{(m)} = 0.09$).

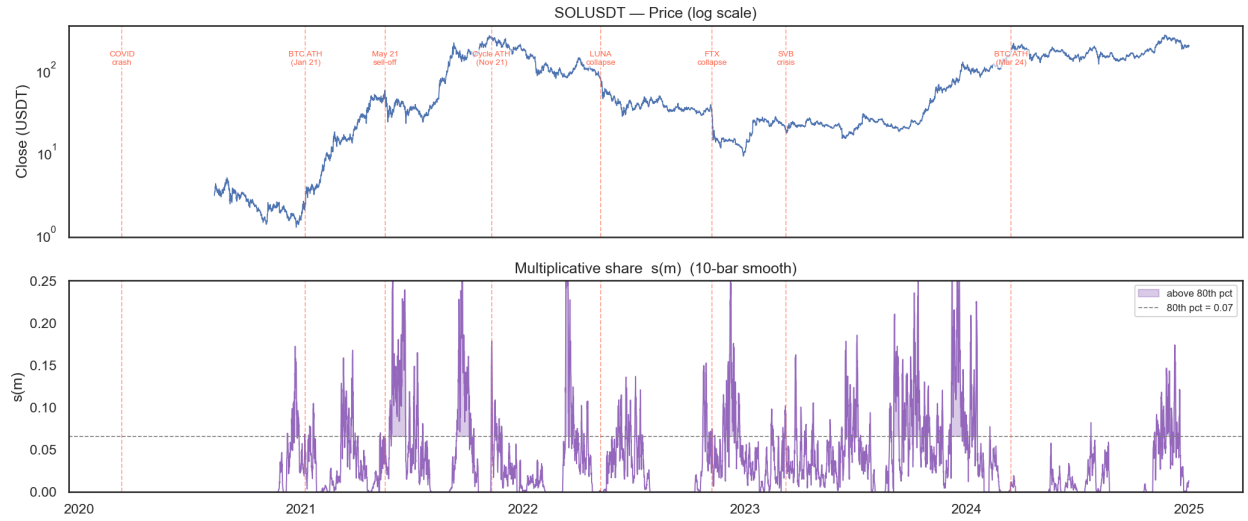


Figure 5: SOLUSDT price (log scale, top) and smoothed multiplicative share $\hat{s}_t^{(m)}$ (10-bar mean, bottom) over the full sample 2020–2024. The dashed line marks the 80th-percentile threshold ($\hat{s}_{0.80}^{(m)} = 0.07$).

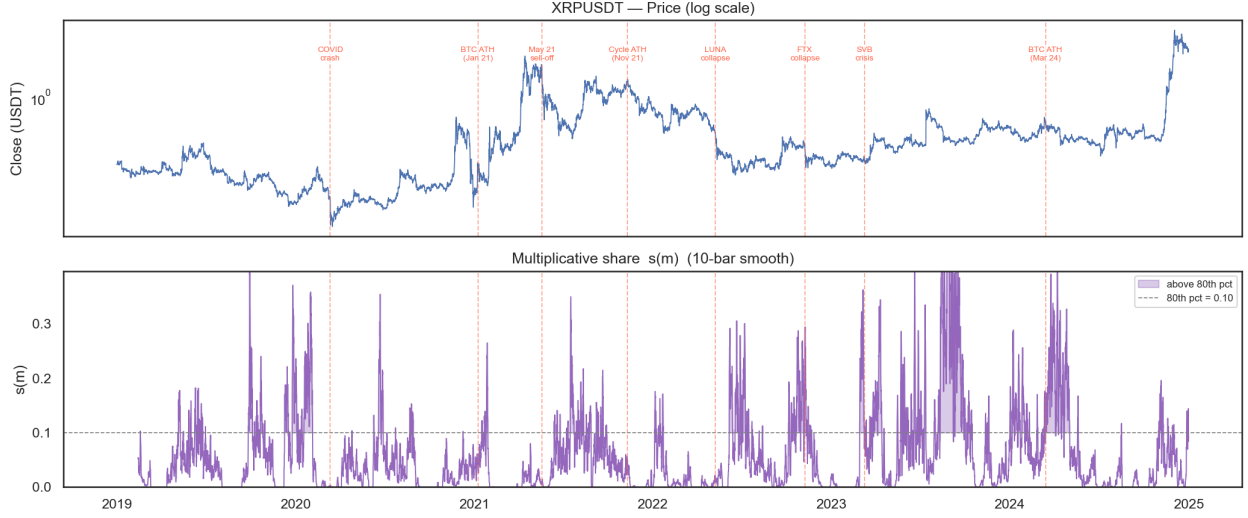


Figure 6: XRPUSDT price (log scale, top) and smoothed multiplicative share $\hat{s}_t^{(m)}$ (10-bar mean, bottom) over the full sample 2019–2024. The dashed line marks the 80th-percentile threshold ($\hat{s}_{0.80}^{(m)} = 0.09$).

Table 3: Episode statistics for the BTCUSDT multiplicative share $\hat{s}_t^{(m)}$ (smoothed, 10-bar mean). The 80th-percentile threshold is 0.067, computed over the full sample. The price return is the log-return over the episode window. max drawdown is the maximum peak-to-trough decline within the window.

Episode	Window	Peak	Peak date	$n_{>80}$	$\%_{>80}$	Ret.	Max DD
Thin liq. era	2019-07 – 2020-02	0.623	2019-07-31	431	29.6%	−22.0%	−40.6%
COVID crash	2020-02 – 2020-05	0.162	2020-02-03	92	12.7%	+0.2%	−51.6%
FTX collapse	2022-10 – 2023-02	0.343	2023-01-10	138	15.2%	+19.2%	−19.0%
Inst. bull run	2024-01 – 2024-12	0.271	2024-09-29	590	26.9%	+121.1%	−8.2%

BTCUSDT — Multiplicative Share $s(m)$ Across Four Market Episodes

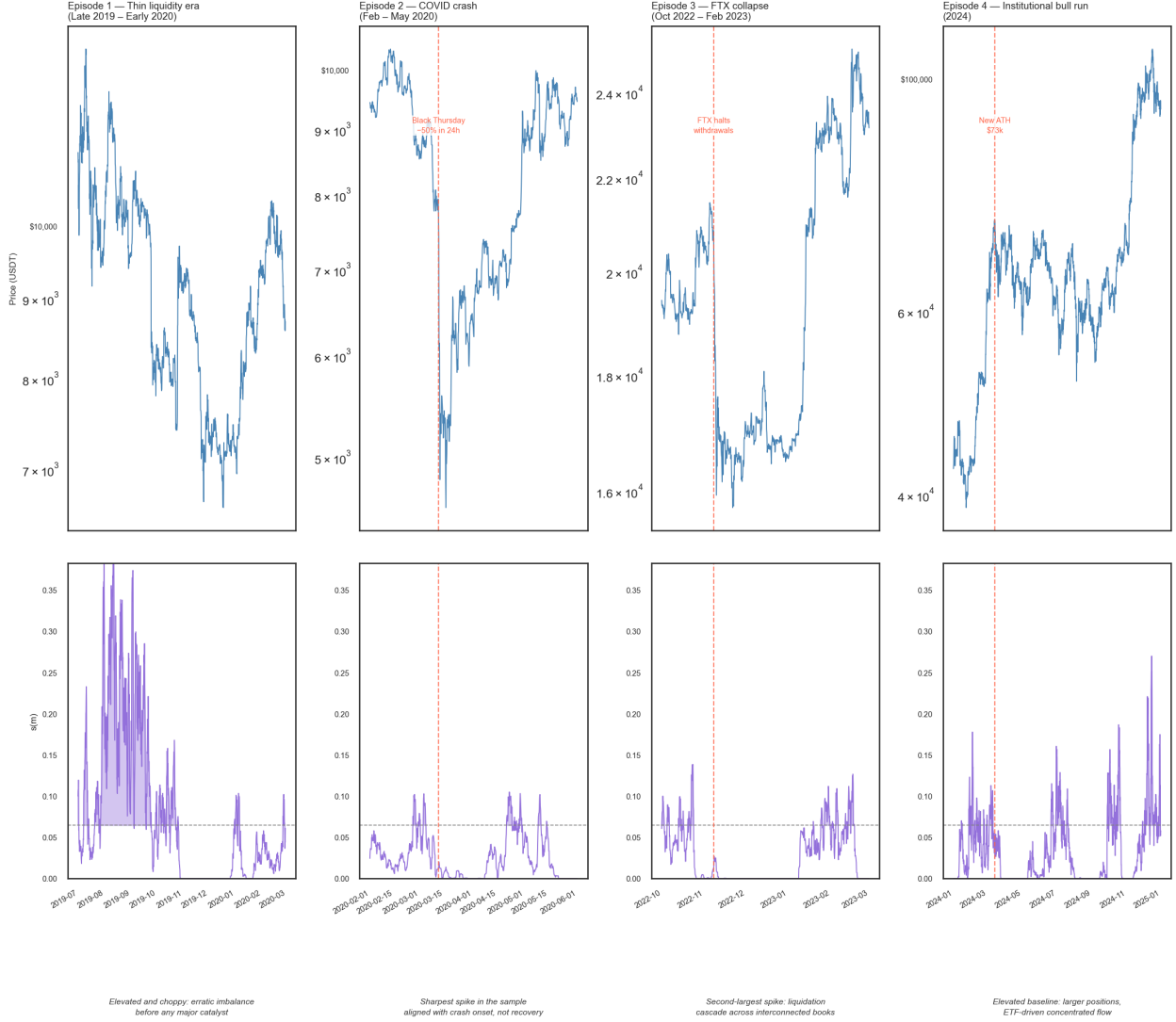


Figure 7: BTCUSDT multiplicative share $\hat{s}_t^{(m)}$ across the four market episodes. Each column shows the BTC price (log scale, top) and the smoothed multiplicative share (10-bar mean, bottom) for a distinct episode: the thin liquidity era (2019 Q3 – 2020 Q1), the COVID crash (Feb–May 2020), the FTX collapse (Oct 2022 – Feb 2023), and institutional bull run (2024). The dashed horizontal line marks the full-sample 80th-percentile threshold. Vertical red dashed lines mark the key event date within each episode. Italic annotations below each share panel describe the economic character of the episodes.

that $\hat{s}_t^{(m)}$ peaks before the volatility event.

Five of the eight events exhibited a negative peak lag. The two strongest cases are the COVID crash (peak lag = -20 bars, = -80 hours, peak correlation 0.360) and the LUNA collapse (peak lag = -18 bars, peak correlation 0.598). Both events are characterized by a well-documented build-up of concentrated, directional order flow in the days preceding the price collapse, which the multiplicative share correctly identifies. The LUNA correlation of 0.598 is the highest in the table and reflects the fact that the LUNA de-pegging mechanism was driven almost entirely by self-reinforcing sell-side OFI, precisely the mechanism the model is designed to capture.

The three events without a negative peak lag — the May 2021 sell-off, the FTX collapse, and the March 2024 ATH — share a common characteristic: each was triggered by a discrete exogenous event (regulatory action, exchange insolvency, and ETF approval, respectively) rather than by a gradual build-up of directional order-flow pressure. In these cases, the multiplicative regime is activated in response to the event rather than in anticipation of it, which explains why contemporaneous or positive-lag correlations dominate.

Table 4: Lead-lag analysis: cross-correlation between $\hat{s}_t^{(m)}$ (smoothed multiplicative share) and the realised price range $(H_t - L_t)/C_t$ over a ± 20 -bar window centered on each event. The peak lag is the lag at which the cross-correlation is maximized. a negative value indicates that $\hat{s}_t^{(m)}$ peaks *before* the volatility event. Contemp. corr. is the cross-correlation at lag zero. A four-hour bar corresponds to four calendar hours.

Event	Asset	Peak lag (bars)	Peak lag (hours)	Peak corr.	Contemp. corr.
COVID crash	BTC	-20	-80	0.360	-0.122
BTC ATH Jan 2021	BTC	-20	-80	-0.132	-0.388
May 2021 sell-off	BTC	$+11$	$+44$	0.014	-0.060
Cycle ATH Nov 2021	BTC	-6	-24	0.191	0.082
LUNA collapse	BTC	-18	-72	0.598	0.075
FTX collapse	BTC	$+3$	$+12$	0.497	0.424
BTC ATH Mar 2024	BTC	$+9$	$+36$	0.030	0.027
XRP legal resolution	XRP	-11	-44	0.269	-0.194

Cross-asset synchronisation. A natural extension of the single-asset analysis is to ask whether the multiplicative regime is activated simultaneously across assets. At each four-hour bar, we compute the fraction of coins whose smoothed multiplicative share exceeds its own 80th-percentile threshold. Figure 8 plots this synchronization measure alongside the BTC price. Episodes in which three or more of the five assets simultaneously enter the multiplicative regime, corresponding to a synchronization fraction above 0.60, are highlighted in red.

The synchronization measure reveals a clear distinction between systemic and idiosyncratic episodes. The COVID crash and the LUNA collapse generated sustained high-synchronization periods, during which the multiplicative regime activated across the entire cross-section of assets. In contrast, the XRP legal resolution generates an isolated spike confined to XRP alone, with no spillover to the remaining assets. The FTX collapse occupies an intermediate position: synchronization is elevated but does not reach the 0.60 threshold, consistent with the episode being driven primarily by exchange-specific flows rather than a broad deterioration in market-wide order flow conditions. This cross-asset dimension of the multiplicative regime provides a simple, model-derived systemic risk indicator with no free parameters beyond the threshold.

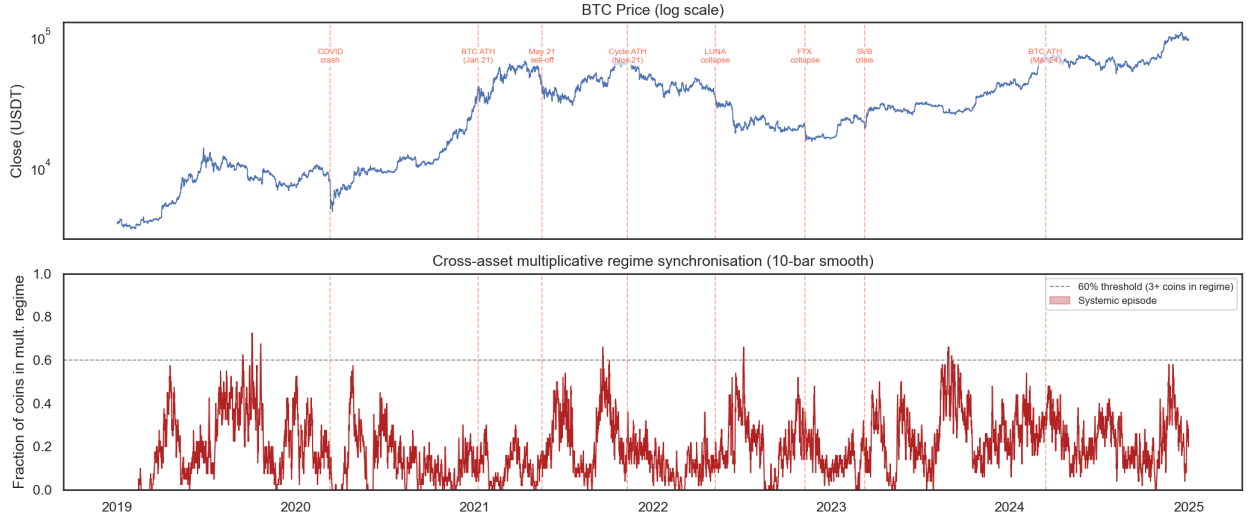


Figure 8: Cross-asset multiplicative regime synchronization. Top panel: BTCUSDT price (logarithmic scale). Bottom panel: fraction of the five coins (BTC, ETH, BNB, SOL, XRP) simultaneously above their own 80th-percentile $\hat{s}^{(m)}$ threshold (10-bar smooth). Red shading marks episodes above 0.60 (three or more coins simultaneously in the multiplicative regime). Vertical dashed lines mark the eight annotated market events.

3.3 Distributional Consequences of the Multiplicative Regime

The previous subsection established *when* the multiplicative regime is activated. We now ask what this implies for the distribution of returns. The theoretical model predicts that when the multiplicative channel dominates, the conditional variance of the OFI increases with the current imbalance state, and through the price-impact equation, this elevated variance transmits into returns. Therefore, the distributional predictions are twofold: returns should exhibit higher variance in the multiplicative regime, and the shape of the return distribution should shift in a manner consistent with state-dependent amplification. We test these predictions using three complementary analyses: a regime-conditional comparison of return moments, an episode-level decomposition of return variance, and a formal assessment of the OFI-induced return variance share $\hat{s}_t^{(m,r)}$.

Regime-conditional return distributions. We partition the full sample of four-hour log-returns into two subsets according to the contemporaneous value of the smoothed multiplicative share: the *additive regime*, defined as bars in which $\hat{s}_t^{(m)} \leq \hat{s}_{0.80}^{(m)}$, and the *multiplicative regime*, defined as bars in which $\hat{s}_t^{(m)} > \hat{s}_{0.80}^{(m)}$. For each asset, we compute the standard deviation, excess kurtosis, and skewness of log-returns within each regime and test for equality of variances across regimes using the Levene test (Levene, 1960).

Table 5 reports these results. The standard deviation is higher in the multiplicative regime for every asset: the increase ranges from a modest 0.5 percent for XRP to 16.2 percent for BTC. The Levene test rejects equal variances at the five percent level for BTC, ETH, BNB, and SOL, confirming that the variance shift is statistically significant and not an artifact of the threshold choice. XRP does not reject the Levene null ($p = 0.311$), consistent with its idiosyncratic dynamics discussed in Section 3.7.

The kurtosis result is more striking and requires careful interpretation than the skewness result. Excess kurtosis is substantially *lower* in the multiplicative regime for every asset: the reduction ranges from 6.80 for XRP to 14.76 for BTC in the full sample. This is consistent with the model. In the additive regime, extreme return events are rare relative to the quiet baseline, which mechanically produces a very high kurtosis. In the multiplicative regime, the elevated variance baseline makes large returns less surprising relative to the prevailing level of order flow activity, compressing the kurtosis even as the absolute magnitude of returns increases. Figure 9 illustrates this: the multiplicative distribution is visibly wider than the additive distribution, but the additive distribution has more pronounced tail outliers relative to its center. This pattern is consistent for all five assets.

Table 5: Regime-conditional return moments. Returns are partitioned into the additive regime ($\hat{s}_t^{(m)} \leq \hat{s}_{0.80}^{(m)}$) and the multiplicative regime ($\hat{s}_t^{(m)} > \hat{s}_{0.80}^{(m)}$). Std is the standard deviation of log-returns in basis points. Kurt is excess kurtosis. Skew is the skewness coefficient. Levene p is the p -value of the Levene test for equality of variances (Levene, 1960). Significance at the five percent level is indicated by *.

Asset	Std (bps)		Excess kurtosis		Skewness		Levene	
	Add.	Mult.	Add.	Mult.	Add.	Mult.	p -val	Sig.
BTC	132.7	154.2	22.01	7.25	-0.75	-0.50	0.000	*
ETH	169.7	178.9	14.23	8.99	-0.73	-0.93	0.020	*
BNB	180.4	183.6	18.51	8.47	-0.27	-0.72	0.009	*
SOL	270.7	290.4	11.20	4.59	0.30	-0.03	0.000	*
XRP	208.1	209.0	28.80	22.00	0.17	0.02	0.311	

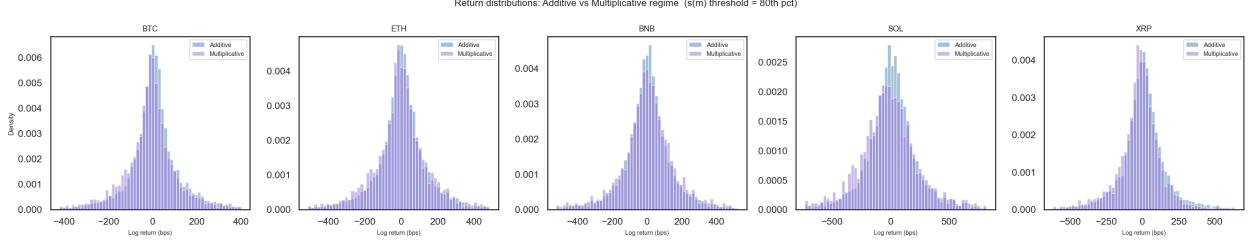


Figure 9: Return distributions conditional on the multiplicative regime. Each panel shows the empirical density of four-hour log-returns (in basis points) for the additive regime ($\hat{s}_t^{(m)} \leq \hat{s}_{0.80}^{(m)}$, blue) and the multiplicative regime ($\hat{s}_t^{(m)} > \hat{s}_{0.80}^{(m)}$, purple). Both distributions are truncated at the 1st and 99th percentiles of the pooled sample for visual clarity.

Episode-level variance decomposition. The regime-conditional analysis pools all multiplicative regime bars across the full sample. To assess whether the OFI channel’s contribution to return variance varies systematically across structural episodes, Table 6 reports an episode-level decomposition for BTCUSDT. For each of the four episodes identified in Section 3.2, we regress realized squared returns r_t^2 on the OFI-induced variance channel $\hat{\beta}_t^2(\hat{\sigma}_a^2 + \hat{\sigma}_m^2 I_t^2)$ within the episode window, and report the resulting R^2 as a measure of the OFI channel’s explanatory power. We also report the ratio $\hat{\beta}^2 \hat{\sigma}_m^2 / \hat{\beta}^2 \hat{\sigma}_a^2$, which measures the relative contribution of the multiplicative to the additive component within the OFI channel.

The results reveal a striking episode-level heterogeneity. During the FTX collapse, the OFI channel explains 10.6 percent of the realized squared returns, approximately ten times the explanatory power achieved in any other episode. The multiplicative-to-additive ratio of 52.1 confirms that during this episode, the OFI variance channel was almost entirely driven by the multiplicative component: the self-amplifying properties of order flow dominated to a degree not observed elsewhere in the sample. During the thin liquidity era, the ratio is 10.2, indicating that multiplicative dynamics were structurally present but spread over a long window with lower total return variance, explaining the low R^2 of 0.004. The COVID crash and the 2024 bull run yield R^2 values below 0.001 suggesting that during these episodes, the OFI channel, while active at the process level, was not the primary driver of the realized magnitude of returns.

The OFI-induced multiplicative return-variance share. The episode-level decomposition aggregates data over several months. To recover the bar-by-bar dynamics of the OFI-to-return transmission, we compute the OFI-induced multiplicative return-variance share $\hat{s}_t^{(m,r)}$ defined in Equation (28), which measures the fraction of the forecast next-bar return variance attributable to the multiplicative OFI channel at each point in time.

Figures 10 and 11 report the full-sample dynamics and the four-episode decomposition for BTCUSDT, respectively. The most important feature of the $\hat{s}_t^{(m,r)}$ series is its dissociation from

Table 6: Episode-level OFI return variance decomposition for BTCUSDT. For each episode, realised squared returns r_t^2 are regressed on the OFI-induced variance channel $\hat{\beta}_t^2(\hat{\sigma}_a^2 + \hat{\sigma}_m^2 I_t^2)$ within the episode window. R^2 measures the fraction of squared-return variation explained by the OFI channel. The ratio $\hat{\beta}^2 \hat{\sigma}_m^2 / \hat{\beta}^2 \hat{\sigma}_a^2$ measures the relative size of the multiplicative versus additive component within the OFI channel. Total ret. var. is the realised variance of log-returns in the episode (bps²).

Episode	Window	Total ret. var.	R^2 (OFI $\rightarrow r^2$)	$\hat{\beta}^2 \hat{\sigma}_m^2 / \hat{\beta}^2 \hat{\sigma}_a^2$
Thin liquidity era	2019-07 – 2020-02	18,763	0.004	10.20
COVID crash	2020-02 – 2020-05	39,844	0.001	3.56
FTX collapse	2022-10 – 2023-02	8,403	0.106	52.15
Institutional bull run	2024-01 – 2024-12	12,288	0.000	7.86

$\hat{s}_t^{(m)}$ in the 2024 episode. Whereas $\hat{s}_t^{(m)}$ is persistently elevated during the institutional bull run, indicating that the OFI process itself is in a multiplicative regime, $\hat{s}_t^{(m,r)}$ remains compressed near zero throughout this episode. This is because the rolling price-impact coefficient $\hat{\beta}_t$ collapsed to near zero in the aftermath of the FTX crisis and remained suppressed throughout 2024. Since $\hat{s}_t^{(m,r)}$ scales with $\hat{\beta}_t^2$, a near-zero beta dampens the transmission of multiplicative OFI dynamics into return variance, regardless of the intensity of the OFI regime itself.

This dissociation is the most important distributional finding of this study. It establishes that the multiplicative OFI channel and the return variance channel are distinct: the former describes the dynamics of the order-flow process in isolation, while the latter measures how those dynamics transmit into prices. The transmission is gated by $\hat{\beta}_t$, which is time-varying and responds to changes in the market structure. During the FTX episode, both $\hat{s}_t^{(m)}$ and $\hat{\beta}_t^2$ were simultaneously large, producing the highest $\hat{s}_t^{(m,r)}$ values in the sample and explaining the episode’s exceptional R^2 in Table 6. During the 2024 bull run, $\hat{s}_t^{(m)}$ was elevated, but $\hat{\beta}_t^2$ was not, decoupling the OFI process from return variance.

3.4 The Price-Impact Link: Rolling β and Return Variance Transmission

The distributional analysis in Section 3.3 establishes that the multiplicative OFI regime is associated with a structurally different return distribution. The present subsection traces the mechanism through which this association operates: the time-varying price-impact coefficient $\hat{\beta}_t$, which determines how strongly a given level of OFI imbalance transmits into contemporaneous returns. We first document the time-series behavior of $\hat{\beta}_t$ and its dependence on the market phase. We then examine the predictive content of the OFI-induced return-variance share $\hat{s}_t^{(m,r)}$ for the realized return variance, completing the empirical chain from the OFI process to return dynamics.

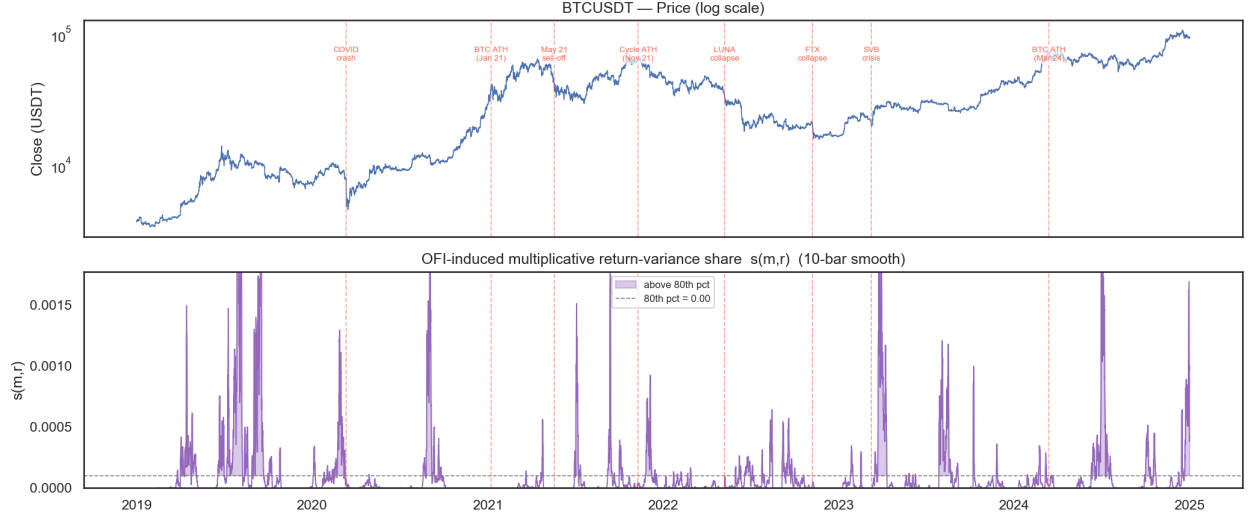


Figure 10: BTCUSDT price (log scale, top) and the OFI-induced multiplicative return-variance share $\hat{s}_t^{(m,r)}$ (10-bar smooth, bottom) over the full sample 2019–2024. The dashed line marks the 80th-percentile threshold of $\hat{s}_t^{(m,r)}$. Shaded regions indicate bars above the threshold. Vertical dashed lines mark the eight annotated market events. Contrast with Figure 2: the 2024 episode shows elevated $\hat{s}_t^{(m)}$ but suppressed $\hat{s}_t^{(m,r)}$, reflecting the compression of the price-impact coefficient $\hat{\beta}_t$ after FTX.

The rolling price-impact coefficient. For each asset we estimate $\hat{\beta}_t$ using a 250-bar rolling OLS regression of the raw four-hour log-return r_{t+1} on the lagged raw relative OFI I_t ,

$$r_{i,t+1} = \hat{\alpha}_t + \hat{\beta}_t I_{i,t} + \hat{\varepsilon}_{i,t+1}, \quad (31)$$

where the intercept $\hat{\alpha}_t$ absorbs the time-of-day level effect within each window, rendering explicit diurnal demeaning as unnecessary. The rolling window corresponds to approximately six weeks of four-hour bars, which is short enough to track structural changes in market conditions while providing adequate statistical accuracy.

Figures 12–16 plot the smoothed rolling $\hat{\beta}_t$ along with the price series for each asset. Several features are common to all assets. First, $\hat{\beta}_t$ is not constant over time; it oscillates between large positive and large negative values, with the amplitude of oscillations increasing substantially during periods of market stress. Second, the most volatile period of $\hat{\beta}_t$ for BTC, ETH, BNB, and SOL coincides with the interval between the LUNA collapse (May 2022) and the FTX aftermath (early 2023), during which $\hat{\beta}_t$ swings between approximately $-1,200$ and $+1,600$ basis points per unit of OFI for BTC and between $-3,000$ and $+1,000$ basis points for BNB. Third, after the FTX crisis, $\hat{\beta}_t$ compresses markedly toward zero for all assets except XRP, remaining close to zero throughout 2024.

BTCUSDT — OFI-Induced Multiplicative Return-Variance Share $s(m,r)$ Across Four Episodes

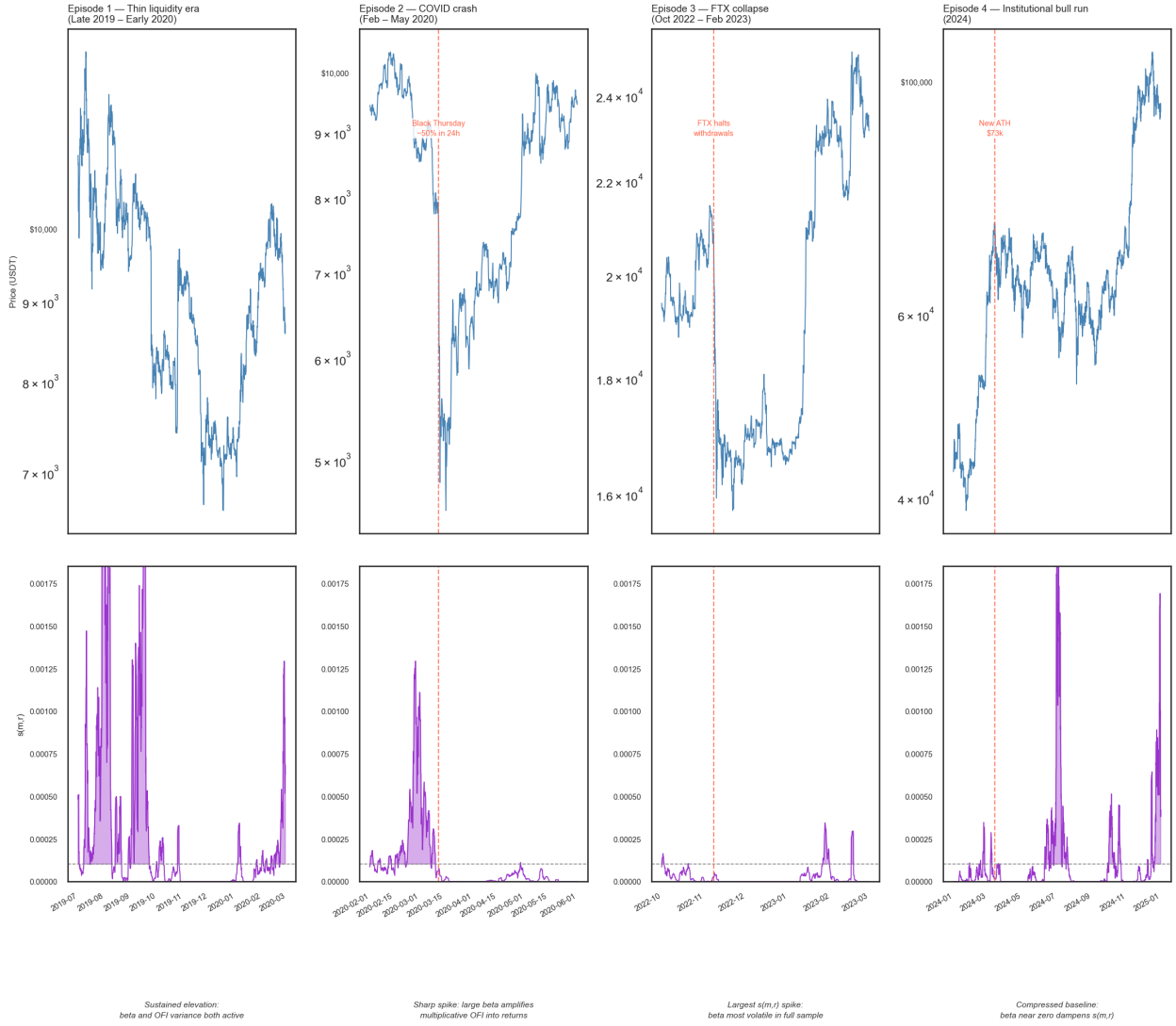


Figure 11: BTCUSDT OFI-induced multiplicative return-variance share $\hat{s}_t^{(m,r)}$ across the same four episodes as Figure 7. The colour is darker (dark orchid) to distinguish $\hat{s}_t^{(m,r)}$ from $\hat{s}_t^{(m)}$ (medium purple) when the two figures are displayed side by side. Episode 3 (FTX collapse) exhibits the largest $\hat{s}_t^{(m,r)}$ values, consistent with the simultaneous elevation of $\hat{s}_t^{(m)}$ and $\hat{\beta}_t^2$ during that period. Episode 4 (institutional bull run) shows compressed $\hat{s}_t^{(m,r)}$ despite elevated $\hat{s}_t^{(m)}$, reflecting the suppression of $\hat{\beta}_t$ after the FTX crisis.

The post-FTX compression of $\hat{\beta}_t$ is the most economically significant time-series feature in the data and is directly connected to the distributional finding of Section 3.3. When $\hat{\beta}_t \approx 0$, the OFI channel transmits essentially no information into returns: even if the OFI process is in a multiplicative regime (elevated $\hat{s}_t^{(m)}$), the return variance channel $\hat{s}_t^{(m,r)}$ remains suppressed because it is scaled by $\hat{\beta}_t^2$. The 2024 bull run therefore represents a market in which large, institutionally driven order flows generated self-amplifying OFI dynamics without a commensurate impact on price formation — a structural decoupling between microstructure and price that has no clear precedent in the earlier part of the sample.

XRP is an exception to this pattern. Its rolling $\hat{\beta}_t$ is positive and significant throughout most of the sample, reaching a peak of approximately +3,500 basis points immediately after the SEC lawsuit announcement in December 2020. During that period, the near-total withdrawal of liquidity from XRP markets meant that any residual buy-side OFI had an outsized positive price impact, consistent with a market in which the effective spread had widened to an extreme level.

Price-impact beta by market phase. To characterize the dependence of $\hat{\beta}_t$ on market conditions in a more structured way, we classify each four-hour bar into one of four market phases using BTC price dynamics: *Bull* (price above the 200-bar rolling mean and rising), *Bear* (price below the rolling mean and falling), *Crash* (drawdown from the rolling peak exceeding 30 percent), and *Recovery* (all remaining bars). For each asset and phase, we report the mean $\hat{\beta}_t$ and the associated t -statistic in Table 7.

The most striking result is the BTC crash-phase coefficient: $\bar{\hat{\beta}}_{\text{Crash}} = -197.5$ basis points with $t = -16.95$, the largest absolute t -statistic in the table. During crashes, buy-side OFI predicts *negative* next-bar returns for BTC returns. This is consistent with a market in which informed sellers use incoming buy-side liquidity to unwind positions: a burst of buy-initiated flow is absorbed by sellers, who then push prices lower, reversing the usual positive OFI-return relationship. ETH exhibits a qualitatively similar pattern, with a crash-phase coefficient of -78.9 ($t = -3.27$).

SOL presents the single most anomalous result in the table: $\bar{\hat{\beta}}_{\text{Crash}} = +137.9$ ($t = +6.61$), which is positive and highly significant. During the crash phase, which primarily corresponds to the LUNA and FTX episodes for SOL, the near-total withdrawal of liquidity from the SOL order book meant that any surviving buy-side OFI moved prices upward with extreme force. This is the microstructural signature of a market under existential stress: the few remaining buyers had unlimited price impact because the sell side had effectively disappeared.

XRP is the only asset with a consistently positive $\hat{\beta}_t$ across all four phases, with Bull and Recovery t -statistics of 9.10 and 11.69 respectively. This reflects XRP’s momentum-driven microstructure: the

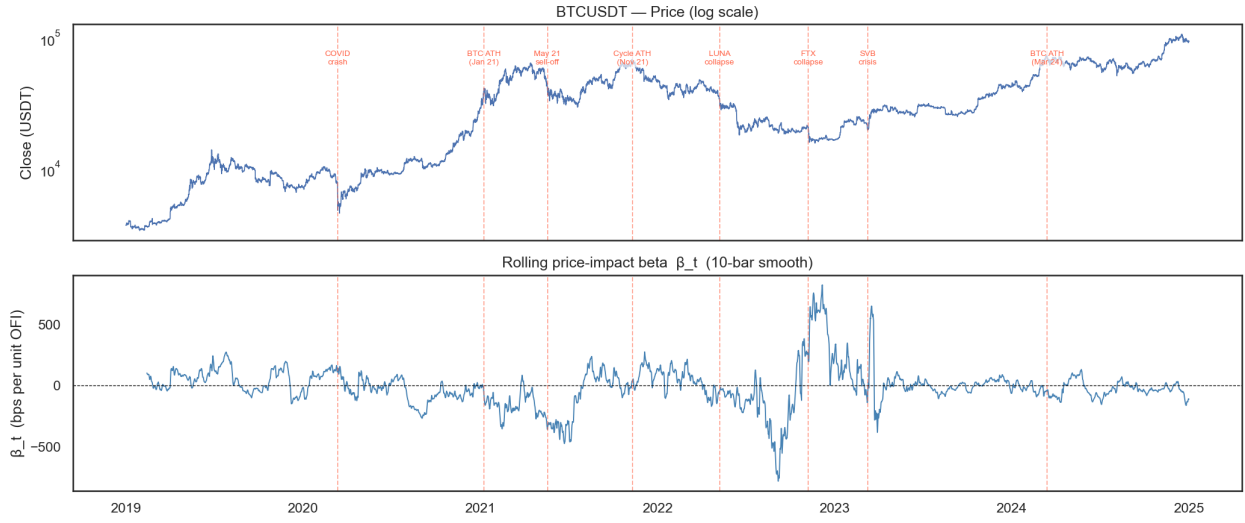


Figure 12: BTCUSDT price (log scale, top) and smoothed rolling price-impact coefficient $\hat{\beta}_t$ (10-bar mean, bottom) over the full sample 2019–2024. $\hat{\beta}_t$ is estimated from a 250-bar rolling OLS regression of r_{t+1} on I_t (equation (31)), reported in basis points per unit of relative OFI. The horizontal dashed line marks $\hat{\beta}_t = 0$. Vertical dashed lines mark the eight annotated market events.

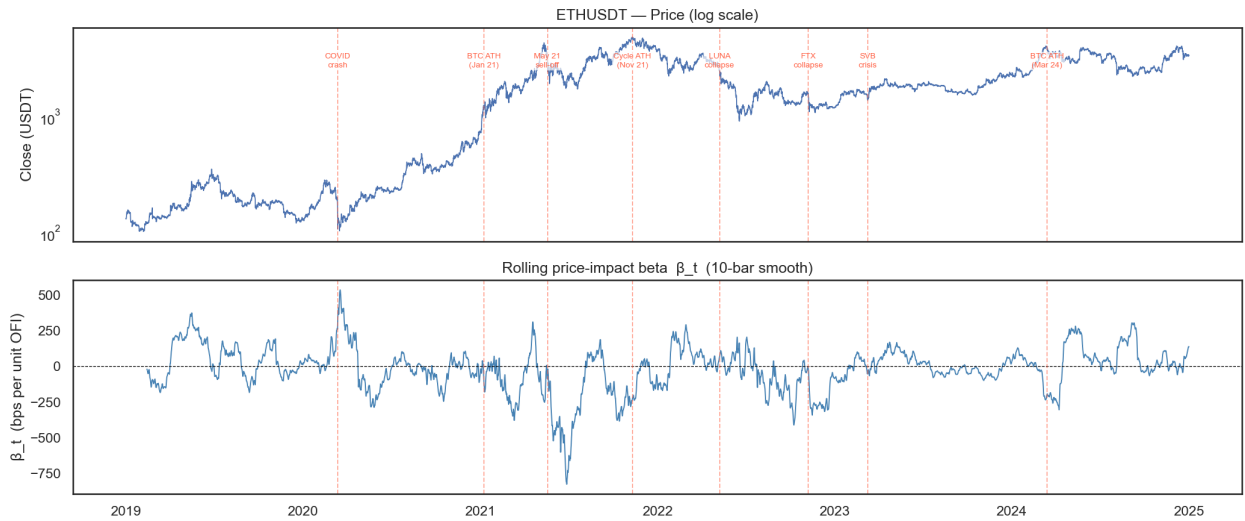


Figure 13: ETHUSDT price (log scale, top) and smoothed rolling price-impact coefficient $\hat{\beta}_t$ (10-bar mean, bottom). See Figure 12 for details.

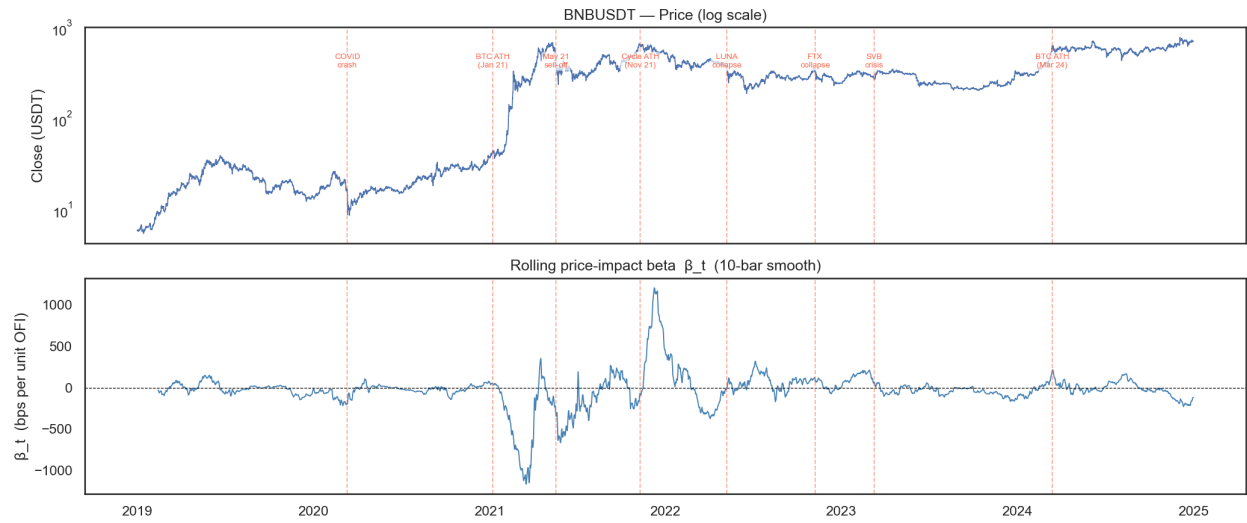


Figure 14: BNBUSDT price (log scale, top) and smoothed rolling price-impact coefficient $\hat{\beta}_t$ (10-bar mean, bottom). BNB exhibits the largest $\hat{\beta}_t$ oscillations in the sample, consistent with its more concentrated ownership structure and the direct influence of Binance exchange events on its order flow. See Figure 12 for details.

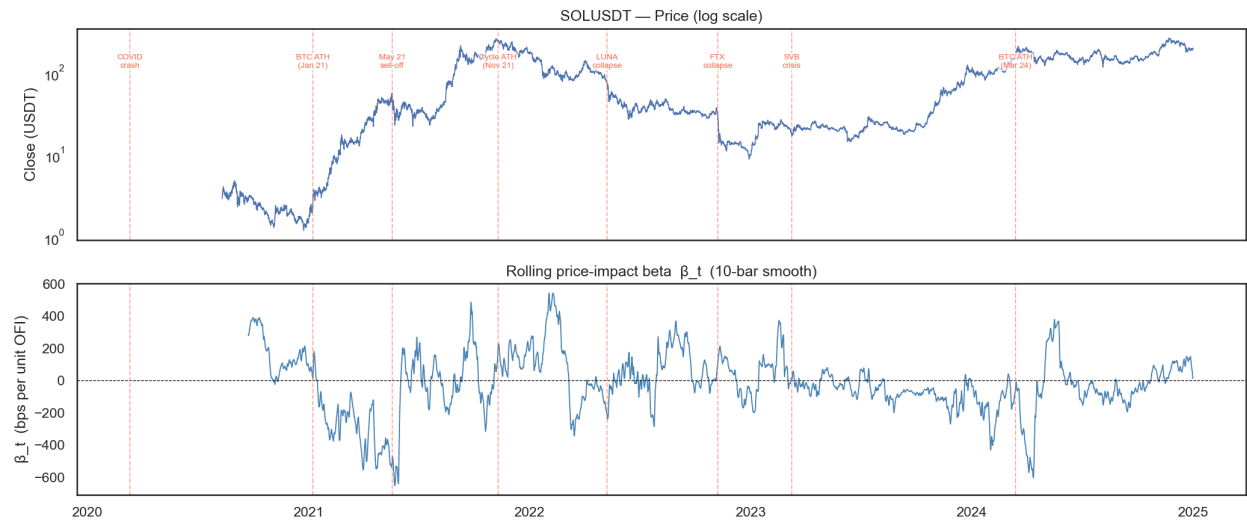


Figure 15: SOLUSDT price (log scale, top) and smoothed rolling price-impact coefficient $\hat{\beta}_t$ (10-bar mean, bottom). The structural break at the cycle ATH (November 2021) is visible: $\hat{\beta}_t$ transitions from large positive values during the bull run to large negative values during the LUNA–FTX bear market. See Figure 12 for details.

buy-side OFI reliably predicts positive next-bar returns regardless of the prevailing market phase, consistent with a market dominated by directional, trend-following participants.

Table 7: Rolling price-impact beta $\hat{\beta}_t$ by market phase. Each four-hour bar is classified as Bull (price above 200-bar rolling mean and rising), Bear (price below rolling mean and falling), Crash (drawdown from rolling peak > 30%), or Recovery (otherwise), using BTC price dynamics. Reported values are the within-phase mean of the rolling $\hat{\beta}_t$ in basis points per unit of relative OFI, with t -statistics ($t = \hat{\beta}/(\hat{s}/\sqrt{n})$) in parentheses.

Asset	Bull		Bear		Crash		Recovery	
	$\hat{\beta}$	t	$\hat{\beta}$	t	$\hat{\beta}$	t	$\hat{\beta}$	t
BTC	-11.3	(-3.00)	2.7	(0.47)	-197.5	(-16.95)	-11.8	(-3.51)
ETH	-38.0	(-10.77)	-10.3	(-1.76)	-78.9	(-3.27)	-27.6	(-8.14)
BNB	-64.5	(-12.74)	13.6	(2.35)	-13.9	(-0.71)	-31.5	(-7.53)
SOL	-51.5	(-7.91)	-13.0	(-1.66)	137.9	(6.61)	-32.9	(-6.19)
XRP	58.0	(9.10)	45.8	(7.69)	-74.0	(-3.48)	53.7	(11.69)

Predictive content of the return-variance share. We now assess whether the OFI-induced multiplicative return-variance share $\hat{s}_t^{(m,r)}$, defined in equation (28), carries forecasting content for the next-bar realized return variance. We estimate the regression

$$r_{i,t+1}^2 = a_i + b_i \hat{s}_{i,t}^{(m,r)} + c_i r_{i,t}^2 + d_i |I_{i,t}| + \eta_{i,t+1}, \quad (32)$$

where $r_{i,t+1}^2$ is the squared four-hour log-return at $t + 1$, $r_{i,t}^2$ is the lagged squared return (a standard GARCH-type control), and $|I_{i,t}|$ is the absolute level of raw OFI (controlling for the direct effect of imbalance magnitude on return variance, independent of the multiplicative channel). The baseline model includes only lagged squared returns. The standard errors are HC3-robust throughout.

Table 8 reports these results. The coefficient $\hat{b}_i$ is negative and significant at the five percent level for BTC ($t = -2.97$, $p = 0.003$), BNB ($t = -3.86$, $p < 0.001$), and XRP ($t = -3.76$, $p < 0.001$). ETH and SOL did not reach the conventional significance levels. The incremental R^2 (ΔR^2) is small in absolute terms, ranging from 0.000 for SOL to 0.003 for XRP, indicating that the forecasting content of $\hat{s}_t^{(m,r)}$ is statistically detectable but economically modest at the four-hour horizon.

The negative sign of $\hat{b}_i$ for BTC, BNB, and XRP reflects the role of market depth as a conditioning variable in the model. In the typical four-hour bar, market depth is sufficient to absorb the concentrated order flow that drives $\hat{s}_t^{(m,r)}$: the multiplicative OFI regime is present at the process level but is quarantined from prices by liquidity provision, causing the system to revert rapidly, and the following bar is calm. This is the depth-absorption mechanism described in Section 2.4:

when $|\hat{\beta}_t|$ is small, as it is in the large majority of bars, and persistently so in the post-FTX period, even elevated $\hat{s}_t^{(m)}$ does not transmit into return variance, and the predictive regression detects the subsequent calm rather than an elevation. The exceptions — the COVID crash and the LUNA collapse, documented in Section 3.2 — are precisely the episodes in which depth was insufficient: $|\hat{\beta}_t|$ was large, the concentrated flow moved prices, and the multiplicative mechanism operated in full. The depth-conditioning hypothesis is tested directly in Section 3.5 using the Amihud (2002) illiquidity ratio as an empirical proxy for the inverse market depth.

Table 8: Predictive regression for realised return variance (equation (32)). The dependent variable is $r_{i,t+1}^2$ (squared four-hour log-return at $t + 1$). $\hat{b}$ is the coefficient on $\hat{s}_{i,t}^{(m,r)}$. R_{base}^2 is the R^2 of the baseline model (lagged r^2 only); R_{full}^2 adds $\hat{s}_{i,t}^{(m,r)}$ and $|I_{i,t}|$; $\Delta R^2 = R_{\text{full}}^2 - R_{\text{base}}^2$. t -statistics are HC3-robust. Significance: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Asset	$\hat{b}$	$t(\hat{b})$	p	R_{base}^2	R_{full}^2	ΔR^2	Sign
BTC	-8.20×10^5	-2.97	0.003	0.009	0.010	0.001	**
ETH	3.58×10^5	0.25	0.799	0.030	0.032	0.002	
BNB	-1.01×10^6	-3.86	0.000	0.024	0.026	0.002	***
SOL	1.73×10^6	0.43	0.670	0.031	0.031	0.000	
XRP	-2.16×10^6	-3.76	0.000	0.033	0.036	0.003	***

Synthesis. The price impact analysis revealed a two-stage transmission mechanism. In the first stage, the OFI process transitions into the multiplicative regime, captured by $\hat{s}_t^{(m)}$, as the order flow imbalance becomes self-amplifying. In the second stage, this multiplicative OFI dynamics transmit into return variance through the price-impact coefficient $\hat{\beta}_t$, captured by $\hat{s}_t^{(m,r)}$. The two stages are distinct: as the 2024 episode demonstrates, multiplicative OFI can persist for extended periods without generating elevated return variance if the price impact channel is suppressed. Conversely, when both channels are simultaneously active, as during the FTX collapse, the OFI mechanism explains a disproportionately large fraction of the realized return variance. The predictive regression confirms that this transmission has measurable out-of-sample content at the four-hour horizon, although the magnitude is modest, suggesting that the model’s primary contribution is structural interpretation rather than short-horizon forecasting.

3.5 Market Depth and the Depth-Absorption Mechanism

The theoretical argument in Section 2.4 implies that the transmission of multiplicative OFI dynamics into return variance is gated by the market depth. We provide a preliminary empirical investigation of this mechanism using the Amihud (2002) illiquidity ratio as a coarse proxy for the inverse market depth. Given the well-known limitations of the Amihud ratio at high frequencies, which conflates price impact with volatility and is sensitive to volume measurement conventions, the results below should be interpreted as suggestive rather than definitive evidence of the

depth-absorption mechanism.

$$\text{ILLIQ}_{i,t} = \frac{|r_{i,t}|}{\text{QV}_{i,t}}, \quad (33)$$

where $|r_{i,t}|$ is the absolute four-hour log-return in basis points, and $\text{QV}_{i,t}$ is the quote volume over the same interval. A high value of $\text{ILLIQ}_{i,t}$ indicates that a given quantity of traded volume moves prices substantially, a signature of thin depth, while a low value indicates that prices are insensitive to volume, consistent with a deep market.

Rolling correlation between $\hat{\beta}_t$ and depth. The depth-absorption hypothesis predicts that $\hat{\beta}_t$ should be positively correlated with $\text{ILLIQ}_{i,t}$: periods of thin depth should be associated with larger price-impact coefficients. Figure 17 shows the smoothed Amihud ratio alongside the smoothed rolling $|\hat{\beta}_t|$ for each asset over the full sample. The two series move together across all five assets, with the most pronounced co-movement during the 2021–2022 bear market, when depth deteriorated sharply and $|\hat{\beta}_t|$ reached its sample maximum. Table 9 reports the rolling correlation between $\text{ILLIQ}_{i,t}$ and $|\hat{\beta}_t|$, estimated over a 250-bar window. The time-series mean is positive and highly significant for BTC ($\rho = 0.015$, $t = 3.31$), ETH ($\rho = 0.020$, $t = 4.52$), and SOL ($\rho = 0.027$, $t = 5.28$), confirming the depth-absorption prediction for these assets: thinner markets are associated with larger price impact coefficients. BNB presents a systematic reversal: its rolling correlation is negative and significant ($\rho = -0.033$, $t = -5.84$). This reversal is not inconsistent with the theoretical framework. The depth-absorption mechanism assumes that liquidity is supplied competitively, so that thin markets translate directly into a higher price impact. BNB’s order flow is dominated by flows originating on or related to the Binance exchange itself, including token burns, listing events, and large institutional flows, which may concentrate in thin-market episodes precisely because these events reduce competitive liquidity provision on both sides of the book simultaneously. In this setting, the Amihud ratio may measure the consequence of concentrated flow rather than the absence of depth, breaking the mapping between illiquidity and price-impact sensitivity that the mechanism requires. This suggests that the depth-absorption hypothesis applies most cleanly to assets whose liquidity is supplied by competitive, decentralized market makers and that BNB constitutes a scope condition rather than a counterexample.

Depth-conditioning of $\hat{s}_t^{(m,r)}$. The depth-absorption hypothesis further predicts that the predictive content of $\hat{s}_t^{(m,r)}$ for the next-bar return variance should be stronger in thin-depth periods. To test this, we split the sample into thin-depth bars ($\text{ILLIQ}_{i,t} > \text{ILLIQ}_i^{(0.70)}$, the 70th-percentile threshold computed over the full sample) and deep-market bars ($\text{ILLIQ}_{i,t} \leq \text{ILLIQ}_i^{(0.70)}$) and re-estimate the predictive regression (32) separately for each subsample. Table 10 reports these results. The evidence is mixed. ETH exhibits the expected pattern: $\hat{\beta}_i$ is positive in the thin-depth subsample and negative in the deep-market subsample, although neither coefficient individually reaches conventional significance. BNB exhibits the opposite pattern: $\hat{\beta}_i$ is negative in the thin-depth

subsample ($t = -4.75$) and positive in the deep-market subsample ($t = 3.36$), a reversal that is statistically significant but runs counter to the simple depth-conditioning prediction. BTC, SOL, and XRP show negative coefficients in both depth regimes, with no significant difference between the subsamples. Taken together, this section provides suggestive, rather than conclusive, evidence for the depth-absorption mechanism. The rolling correlation results support the mechanism for BTC, ETH, and SOL, the three assets whose liquidity is most likely to be supplied competitively. The depth-split regression results are weaker, consistent with the Amihud ratio being a noisy proxy for effective depth at the four-hour frequency. A more direct test would require order-book-level measures of resting liquidity at multiple price levels, data that are not available for the full 2019–2024 sample at this frequency. Therefore, we treat Section 3.5 as a preliminary characterization of the depth-absorption mechanism and leave a formal test for future work with higher-resolution liquidity data.

Table 9: Rolling correlation between the Amihud illiquidity ratio $ILLIQ_{i,t}$ and the absolute rolling price-impact coefficient $|\hat{\beta}_t|$, estimated over a 250-bar rolling window. The reported value is the time-series mean of the rolling correlation, with the t -statistic computed as mean divided by Newey–West standard error with 20 lags. A positive mean correlation confirms that thinner markets (higher $ILLIQ$) are associated with larger price-impact coefficients.

Asset	Mean rolling corr.	t -statistic
BTC	0.015	3.31
ETH	0.020	4.52
BNB	−0.033	−5.84
SOL	0.027	5.28
XRP	0.003	0.46

Table 10: Predictive regression (32) estimated separately on thin-depth bars ($ILLIQ_{i,t} > ILLIQ_i^{(0.70)}$) and deep-market bars ($ILLIQ_{i,t} \leq ILLIQ_i^{(0.70)}$). $\hat{b}^{\text{thin}}$ and $\hat{b}^{\text{deep}}$ are the coefficients on $\hat{s}_{i,t}^{(m,r)}$ in each subsample. t -statistics are HC3-robust. Significance: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Asset	Thin depth ($> 70\text{th pct}$)			Deep market ($\leq 70\text{th pct}$)		
	$\hat{b}^{\text{thin}}$	t	Sig.	$\hat{b}^{\text{deep}}$	t	Sig.
BTC	-5.07×10^6	−3.35	**	-2.89×10^6	−3.12	**
ETH	3.05×10^7	1.51		-2.08×10^6	−1.08	
BNB	-1.07×10^7	−4.75	***	3.61×10^6	3.36	***
SOL	-1.06×10^7	−0.81		-2.28×10^6	−0.68	
XRP	-6.52×10^5	−0.17		-3.10×10^6	−1.78	

3.6 Timing of Large Price Moves: Does the Multiplicative Regime Lead?

The preceding subsections established that the multiplicative OFI regime is associated with a structurally distinct return distribution, that the price-impact coefficient amplifies this regime into

return variance, and that the regime exhibits leading-indicator properties around specific crisis events. A natural question is whether these leading-indicator properties generalize unconditionally across the full sample: do $\hat{s}_t^{(m)}$ spikes systematically precede large price moves, or are the documented lead-lag relationships confined to the specific episodes examined in Section 3.2? We addressed this question in two steps. First, we test whether the distribution of forward waiting times from spike onsets to large price moves differs from the geometric null that would be obtained under pure randomness. Second, we interpret the results in light of the episodic structure documented earlier.

Setup. A *spike onset* is defined as the first bar of a contiguous run in which the smoothed multiplicative share $\hat{s}_t^{(m)}$ exceeds its 70th-percentile threshold. Using the 70th rather than the 80th percentile increased the number of detected onsets, providing greater statistical power for the tests that followed. A *large price move* is defined as a four-hour log-return satisfying $|r_t| > 2 \hat{\sigma}_t$, where $\hat{\sigma}_t$ is the expanding-window standard deviation of log returns estimated on all bars strictly prior to t , ensuring no look-ahead bias in the threshold. For each spike onset at bar s_i , we record the forward distance $d_i = \min\{k > 0 : s_i + k \text{ is a large price move, } k \leq H\}$, where the horizon $H = 60$ bars ($= 240$ hours ≈ 10 days). Spike onsets for which no large move occurs within the horizon are not counted, making the analysis conservative: we study only those spikes that are eventually followed by a large move and ask how quickly.

The geometric null. If spike onsets carry no predictive content for the timing of large price moves, the forward distance d_i should follow a geometric distribution with parameter p equal to the unconditional large-move arrival rate. This follows from the memoryless property of the Poisson process: given that no large move has yet occurred, the probability of one arriving at the next bar is constant at $\hat{p}$ regardless of the time elapsed since the spike onset. We estimate $\hat{p}$ from the unconditional large-move frequency in the full series, avoiding any circularity with spike detection. The truncated geometric CDF on the support $[1, H]$ then serves as the null distribution against which we test the empirical distribution of d_i .

Tests. We apply two complementary tests to this data. The first is a *Kolmogorov–Smirnov test* (Kolmogorov, 1933; Smirnov, 1948), which evaluates the maximum absolute deviation between the empirical CDF of d_i and the geometric null CDF. To obtain a valid p -value without relying on the standard asymptotic approximation, which assumes independent and identically distributed draws, an assumption that forward distances from spike onsets may violate, we compute the p -value by permutation: we randomly relocate the spike onsets $N = 1,000$ times, compute the KS statistic for each permutation, and report the fraction of permuted KS statistics that exceed the observed value. The second is a *chi-squared goodness-of-fit test* on five equal-width bins covering $[1, H]$, which is more sensitive than the KS test to systematic deviations in specific lag ranges (such as the over-representation at short lags suggested by the visual inspection of histograms). The shuffle p -value from the median comparison in Table 11 provides a third complementary test of

whether the central tendency of the forward-distance distribution differs from the geometric null.

Results. Table 11 reports the forward-distance statistics for all five assets, and Table 12 reports the KS and chi-squared test results for the entire sample. Figure 19 visualizes the empirical density and CDF against the geometric null for all assets.

Several features of Table 11 are noteworthy. The match rate, the fraction of spike onsets followed by a large move within 60 bars, ranges from 44.1 percent for SOL to 77.3 percent for BTC. The observed median forward distance is shorter than the shuffle median for every asset: by two bars for BTC, one bar for ETH and BNB, four bars for SOL, and three bars for XRP. In absolute terms, spike onsets precede large price moves by a median of 48 to 64 hours across the five assets. However, none of the shuffle p -values reaches conventional significance at the five percent level, with the closest being XRP ($p = 0.072$) and SOL ($p = 0.100$) values.

Table 11: Forward-distance statistics: time from $\hat{s}_t^{(m)}$ spike onset (70th percentile threshold) to the next large price move ($|r_t| > 2\hat{\sigma}_t$) within a horizon of $H = 60$ bars (= 240 hours). Match rate is the fraction of spike onsets matched to a large move within the horizon. Median lag is the median forward distance in bars and hours. Shuffle median is the median forward distance under 1,000 random relocations of the spike onsets. p (spike < rand) is the fraction of shuffles with a lower median than observed; $p < 0.05$ indicates that spikes precede large moves faster than chance.

Asset	n_{spikes}	n_{matched}	Match rate	Median lag		Shuffle med.	p (spike < rand)
				(bars)	(hours)		
BTC	154	119	77.3%	13.0	52	15.0	0.156
ETH	148	98	66.2%	14.5	58	15.5	0.355
BNB	238	150	63.0%	16.0	64	17.0	0.362
SOL	118	52	44.1%	15.0	60	19.0	0.100
XRP	265	169	63.8%	12.0	48	15.0	0.072

Table 12: Kolmogorov–Smirnov and chi-squared goodness-of-fit tests of the forward-distance distribution against the geometric null. $\hat{p}_{\text{geom}}$ is the estimated geometric parameter (unconditional large-move arrival rate). KS stat is $\sup_k |F_n(k) - F_{\text{geom}}(k)|$. KS p -val is permutation-based (fraction of 1,000 shuffles with KS stat $\geq$ observed). χ^2 p -val is from the chi-squared goodness-of-fit test on five equal-width bins. * denotes $p < 0.10$; no asset rejects the geometric null at the five percent level.

Asset	n	$\hat{p}_{\text{geom}}$	KS stat	KS p -val	χ^2 stat	χ^2 p -val	Verdict
BTC	119	0.045	0.057	0.877	2.32	0.678	Geometric
ETH	98	0.045	0.082	0.728	1.27	0.866	Geometric
BNB	150	0.038	0.087	0.593	5.49	0.241	Geometric
SOL	52	0.024	0.119	0.419	2.13	0.713	Geometric
XRP	169	0.050	0.101	0.596	9.36	0.053*	Geometric

Interpreting the KS result. The KS and chi-squared tests do not reject the geometric null for any asset, indicating that the *shape* of the forward-distance distribution from spike onsets is consistent with a memoryless process. This is not a failure of the model; it is a precise and important boundary result.

Two interpretations reconcile the non-rejection with the leading-indicator evidence in Section 3.2. The first concerns the nature of the geometric null point. The test compares the empirical distribution to a geometric process with an *unconditional* arrival rate $\hat{p}$. If the effective arrival rate of large moves is elevated in the period following a spike, but not so dramatically that it changes the shape of the distribution, the KS test will not detect this, while the shuffle median comparison will show a consistently shorter forward distance. The data are consistent with this interpretation: the shuffle medians are uniformly higher than the observed medians, but the differences are too small to be significant given the available sample sizes.

The second concern is the selectivity of the leading indicator signal. The lead-lag analysis in Section 3.2 shows that the negative peak-lag result holds strongly for crisis events driven by a gradual build-up of directional order-flow pressure (the COVID crash and the LUNA collapse) but not for events triggered by discrete exogenous shocks (FTX and ETF approval). When all spike-to-move pairs are pooled unconditionally, as in the present analysis, the predictive signal from the former group is diluted by the noise from the latter, producing a distribution that is statistically indistinguishable from the geometric null.

Match rates as a bounding result. The match rates in Table 11 provide a complementary perspective on this. For BTC, 77.3 percent of spike onsets are followed by a 2σ large move within ten days. At first glance, this appears high; however, the unconditional probability of a 2σ move within 60 bars — given $\hat{p}_{\text{geom}} = 0.045$ — is $1 - (1 - 0.045)^{60} \approx 93$ percent. The observed match rate of 77.3 percent is therefore *below* the unconditional expectation: spike onsets are *less* likely to be followed by a large move within 10 days than a randomly chosen bar. This reflects the conservative design of the analysis: spike onsets concentrate in periods of elevated OFI activity, which tend to occur after large moves rather than before. The spike onset definition, requiring the *first* bar of each above-threshold run, ensures that we do not double-count within sustained high- $\hat{s}_t^{(m)}$ periods, but it also means that many spike onsets occur in the wake of an existing large move rather than in advance of one.

Summary. Taken together, these results establish a precise characterization of the predictive content of the multiplicative OFI. The forward-distance distribution from spike onsets to large price moves is consistent with a memoryless geometric process, ruling out any unconditional mechanical relationship between $\hat{s}_t^{(m)}$ spikes and subsequent large returns. The leading-indicator properties documented in Section 3.2 are therefore event-specific: they characterize episodes in which the

multiplicative regime builds gradually before a crisis and do not generalize to an unconditional timing rule. The correct interpretation of $\hat{s}_t^{(m)}$ is not as a short-horizon trigger for large price moves but as a structural state variable that identifies periods of concentrated, self-amplifying order flow — a regime that is associated with distributional shifts, elevated return variance transmission, and specific classes of crisis events, but does not generate a statistically robust unconditional timing signal at the four-hour frequency.

3.7 Cross-Sectional Heterogeneity

The preceding subsections treated the five assets primarily through the lens of BTC, using the remaining coins for corroboration. We now systematically examine the cross-sectional variation in the model’s estimates and empirical findings, asking what drives differences across assets and what these differences reveal about the relationship between market structure and multiplicative OFI dynamics.

Ordering assets by multiplicative intensity. A natural summary of cross-sectional heterogeneity is the Fama–MacBeth estimate of $\hat{\sigma}_m^2$ and its associated t -statistic from Table 2. Ranked by the consistency of the multiplicative component across sub-periods, the assets order as follows: XRP ($t = 5.05$), BNB ($t = 4.14$), SOL ($t = 3.99$), ETH ($t = 3.63$), and BTC ($t = 3.26$). This ordering is economically interpretable and aligns with measurable differences in market structure, ownership concentration, and sources of order-flow pressure across assets.

XRP: regulatory and ownership concentration. XRP exhibits the highest multiplicative t -statistic and the lowest dominance threshold ($\hat{\sigma}_a/\hat{\sigma}_m = 0.227$), meaning that a net directional OFI imbalance of only 22.7 percent suffices to trigger multiplicative dominance. This reflects two structural features that are specific to XRP. First, XRP’s ownership is highly concentrated relative to the other four assets: a small number of wallets control a disproportionate share of the circulating supply, which means that large holder activity generates outsized order-flow pressure relative to the background noise. Second, the XRP sample period includes the SEC lawsuit filed in December 2020 and its resolution in July 2023, events that periodically reduced market participation to a thin core of committed buyers and sellers, amplifying the sensitivity of OFI dynamics to individual large orders. The rolling $\hat{\beta}_t$ for XRP (Figure 16) reflects this idiosyncratic structure: it is predominantly positive and significant throughout all market phases (Table 7), consistent with a market in which the buy-side OFI reliably predicts price appreciation regardless of the broader market environment. The cross-asset synchronization analysis confirms that XRP’s multiplicative regime activations are largely idiosyncratic: the legal resolution episode generated no detectable spillover to the other four assets.

BNB: exchange-driven concentration. BNB has the highest additive variance component ($\hat{\sigma}_a^2 = 0.010$, approximately 2.7 times the BTC estimate) and the highest rolling $\hat{\beta}_t$ oscillations in the sample, reaching approximately $-3,000$ basis points during the 2021 bull market period. Both features are consistent with BNB’s unique position as the native token of the Binance exchange: its order flow is directly affected by exchange-level events — regulatory announcements, listing decisions, token burns, and large institutional flows on the Binance platform — that generate state-independent background noise (elevated $\hat{\sigma}_a^2$) while simultaneously creating concentrated directional pressure ($\hat{\sigma}_m^2 = 0.100$, the highest point estimate among the five assets). The R^2 of the BNB variance equation is 0.020, the highest in Table 2, reflecting that the linear x_t^2 specification fits BNB’s variance dynamics more cleanly than it fits the other assets, possibly because Binance-specific flows are more systematically proportional to the prevailing OFI level than the heterogeneous flows that drive BTC and ETH.

SOL: cycle dynamics and structural breaks. SOL is the youngest asset in the sample, with data available only from mid-2020, and its order flow dynamics are dominated by two structural episodes: the explosive bull market of 2021 and the near-total collapse of the LUNA–FTX period. The rolling $\hat{\beta}_t$ for SOL (Figure 15) shows the clearest structural break of any asset: a transition from large positive coefficients during the bull market ($\tilde{\beta}_{\text{Bull}} = -51.5$, consistent with the mean-reverting character of the bull market phase) to a crash-phase coefficient of $+137.9$ ($t = 6.61$) during LUNA–FTX, the only positive crash-phase coefficient in Table 7. As discussed in Section 3.4, this positive crash-phase beta reflects the near-total withdrawal of liquidity from the SOL order book: the few surviving buyers had essentially unlimited price impact because the sell side had disappeared. The relatively low match rate in Analysis H (44.1 percent) is also connected to this episode structure: during the LUNA–FTX crash, SOL experienced sustained large moves over an extended period, meaning that the forward-distance measure is less informative for this asset than for the others.

ETH: depth and the limits of the linear specification. ETH has the highest dominance threshold (0.430) and the lowest $\hat{\sigma}_m^2$ point estimate (0.033), consistent with its status as the deepest and most liquid market of the five. A larger OFI imbalance is required before the multiplicative channel dominates because a deeper order book absorbs more order flow before self-amplifying dynamics emerge. The R^2 of the variance equation for ETH is 0.002, the lowest in Table 2, suggesting that the linear x_t^2 specification captures less of ETH’s variance dynamics than for the other assets. This is consistent with ETH’s role as the hub of the DeFi ecosystem: its order flow is shaped by a wider variety of participants and mechanisms (liquidations, arbitrage across DeFi protocols, and staking flows) than any other asset in the sample, and the resulting variance structure is correspondingly more complex. The Levene test for ETH in Table 5 rejects equal variances across regimes ($p = 0.020$), confirming that the multiplicative regime is economically meaningful for ETH, even if the linear variance equation provides a less precise fit.

BTC: the benchmark case. BTC has the lowest multiplicative t -statistic (3.26) among the five assets, yet it yields the most interpretable results across the full empirical analysis. The four-episode decomposition (Section 3.2) is clearest for BTC because the BTC market is the most institutionally diverse; it spans retail, hedge fund, and ETF flows, which produce a richer variety of multiplicative regime activations than the more concentrated or idiosyncratic markets of the other assets. The crash-phase beta (-197.5 , $t = -16.95$) is the largest absolute t -statistic in Table 7, and the FTX episode yields the highest OFI-to-return R^2 of any asset in any episode (0.106, as shown in Table 6. The post-FTX compression of $\hat{\beta}_t$ toward zero — the most structurally important finding of Section 3.4 — is also clearest for BTC, where it persists throughout the entire 2024 sample and produces the dissociation between $\hat{s}_t^{(m)}$ and $\hat{s}_t^{(m,r)}$ that defines the institutional bull run episode.

Threshold heterogeneity and market depth. A clean cross-sectional regularity emerges from comparing the dominance thresholds: ETH (0.430) > SOL (0.418) > BTC (0.328) > BNB (0.277) > XRP (0.227). This ordering broadly correlates with market depth: ETH and SOL require the largest imbalance before multiplicative dynamics dominate, consistent with their deeper order books; XRP and BNB require the smallest, consistent with their more concentrated flow structures. The t -statistics on the threshold (Table 2) are high for all five assets (3.38 to 11.87), confirming that the threshold is a stable, well-identified structural parameter in every sub-period, and not merely a sample artifact.

Regime distributions: a cross-sectional anomaly. The only asset for which the Levene test does not reject equal variances across regimes is XRP ($p = 0.311$, Table 5). This is paradoxical given that XRP has the highest multiplicative t -statistic; one might expect the asset with the most consistent multiplicative component to show the clearest regime-conditional variance difference. The resolution lies in the nature of XRP’s multiplicative episodes. XRP’s multiplicative regime activations are concentrated around the legal resolution event of 2023 and the subsequent rally, a period during which return variance was high in both the additive and multiplicative regimes because the legal uncertainty generated elevated volatility even during periods of relatively balanced order flow. The Levene test therefore fails not because the multiplicative OFI dynamics are absent, but because the additive regime itself was unusually volatile during the legal episode, compressing the variance differential between the two regimes.

Summary. Table 13 summarizes the key cross-sectional statistics from the study. The five assets span a range of market structures, from the deep, institutionally diverse BTC market to the concentrated, idiosyncratic XRP market, and the model’s parameters vary systematically with these structural differences. The multiplicative component $\hat{\sigma}_m^2$ is consistently significant across all assets and sub-periods, confirming the universality of the additive multiplicative OFI structure within the cryptocurrency asset class. Simultaneously, the quantitative differences in the threshold, multiplicative intensity, and price-impact dynamics across assets reflect the richness of the cross-

sectional variation and provide a foundation for future work linking OFI dynamics to on-chain measures of ownership concentration and market depth.

Table 13: Cross-sectional summary of key empirical results. $\hat{\sigma}_m^2$ and $t(\hat{\sigma}_m^2)$ are from the Fama–MacBeth estimation (Table 2). Threshold = $\hat{\sigma}_a/\hat{\sigma}_m$ (FM mean). Var. ratio is $\text{Std}_{\text{Mult}}/\text{Std}_{\text{Add}}$ from Table 5. Crash $\hat{\beta}$ and its t -statistic are from Table 7. FTX R^2 is the episode-level OFI-to-return R^2 from Table 6. Levene sig. indicates whether the Levene test rejects equal variances across regimes at the five percent level.

Asset	$\hat{\sigma}_m^2$	$t(\hat{\sigma}_m^2)$	Threshold	Var. ratio	Crash $\hat{\beta}$	$t(\text{crash})$	FTX R^2	Levene
BTC	0.082	3.26	0.328	1.16	−197.5	−16.95	0.106	✓
ETH	0.033	3.63	0.430	1.05	−78.9	−3.27	—	✓
BNB	0.100	4.14	0.277	1.02	−13.9	−0.71	—	✓
SOL	0.061	3.99	0.418	1.07	137.9	6.61	—	✓
XRP	0.101	5.05	0.227	1.00	−74.0	−3.48	—	

4 Conclusion

This study models order-flow imbalance in cryptocurrency markets as an additive multiplicative stochastic process and derives a structural decomposition of conditional return variance into an additive and multiplicative OFI channel. The framework provides a direct mechanism through which the statistical properties of order flow — its heavy tails, regime-dependence, and sensitivity to the presence of large, concentrated participants — transmit into the distribution of asset returns.

The central empirical result is that the multiplicative variance component $\hat{\sigma}_m^2$ is positive and statistically significant for all five assets across 12 independent six-month sub-periods, with Fama–MacBeth t -statistics ranging from 3.26 for BTC to 5.05 for XRP. This finding establishes that state-dependent OFI amplification is a persistent structural feature of cryptocurrency order flow and not a consequence of any single crisis episode. The dominance threshold $\hat{\sigma}_a/\hat{\sigma}_m$, which ranges from 0.227 for XRP to 0.430 for ETH, provides a directly interpretable measure of the imbalance magnitude required before the multiplicative channel begins to dominate the market. Its cross-sectional ordering correlates with observable differences in market depth, ownership concentration, and sources of order flow pressure across assets.

The multiplicative share $\hat{s}_t^{(m)}$ is activated episodically rather than continuously. Around crisis events driven by a gradual build-up of directional order-flow pressure, such as the COVID crash of March 2020 and the LUNA collapse of May 2022, the indicator leads realized volatility by up to 80 hours, providing evidence that the model detects structural fragility before it manifests in prices. Around events triggered by discrete exogenous shocks — the FTX collapse and the ETF-driven 2024 bull run — the multiplicative regime activates in response rather than in anticipation, and the

leading-indicator property does not hold. The Kolmogorov–Smirnov test against the geometric null formalizes this distinction: the unconditional forward-distance distribution from multiplicative regime onsets to large price moves is consistent with a memoryless process, confirming that the model’s predictive content is episode-specific rather than a general timing signal.

Whether multiplicative OFI dynamics transmit into return variance depends critically on market depth. The price-impact coefficient $\hat{\beta}_t$ is positively correlated with the Amihud illiquidity ratio for BTC, ETH, and SOL, confirming that thinner markets exhibit larger price impact coefficients. During the FTX collapse, both the multiplicative share and the price-impact coefficient were simultaneously elevated, and the OFI channel explained 10.6 percent of realized squared returns, approximately ten times its explanatory power in any other episode. During the 2024 institutional bull run, the multiplicative OFI regime was persistently active, but $\hat{\beta}_t$ collapsed to near zero following the structural shock of FTX, quarantining the multiplicative channel from prices. This dissociation between the OFI process and the return variance channel is the most practically important finding of this study: the presence of concentrated, self-amplifying order flow is a necessary but not sufficient condition for large return realizations.

Several directions for future research emerge from these findings. First, the cross-sectional ordering of the dominance threshold invites a direct empirical link to on-chain measures of ownership concentration — Gini coefficients of wallet wealth, Pareto exponents, and large-holder activity ratios — which are observable in cryptocurrency markets in a way that has no analogue in traditional asset classes. Second, the depth-absorption mechanism documented here calls for a more direct empirical test than the Amihud ratio can provide at the four-hour frequency. Re-estimating the mechanism using order-book-level measures of resting liquidity, such as aggregate quoted depth within five basis points of the midpoint, would transform the preliminary evidence of Section 3.5 into a formal test of the theoretical prediction and clarify whether the BNB reversal reflects a genuine scope condition or a measurement artifact. Third, extending the additive multiplicative framework to a multivariate setting — modelling the joint OFI process across assets and allowing for cross-asset multiplicative spillovers — would provide a formal foundation for the cross-asset synchronisation indicator introduced in Section 3.2 and connect the model to the systemic risk literature.

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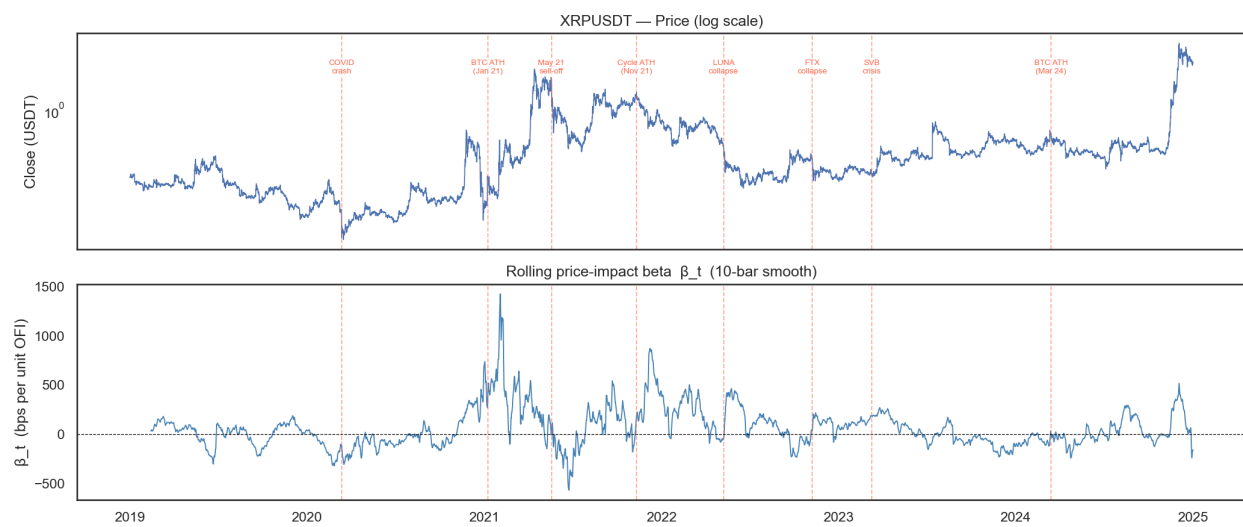


Figure 16: XRPUSD price (log scale, top) and smoothed rolling price-impact coefficient $\hat{\beta}_t$ (10-bar mean, bottom). Unlike the other four assets, XRP maintains a predominantly positive $\hat{\beta}_t$ throughout the sample, with the largest spike (approximately +3,500 bps) immediately following the SEC lawsuit announcement in December 2020. See Figure 12 for details.

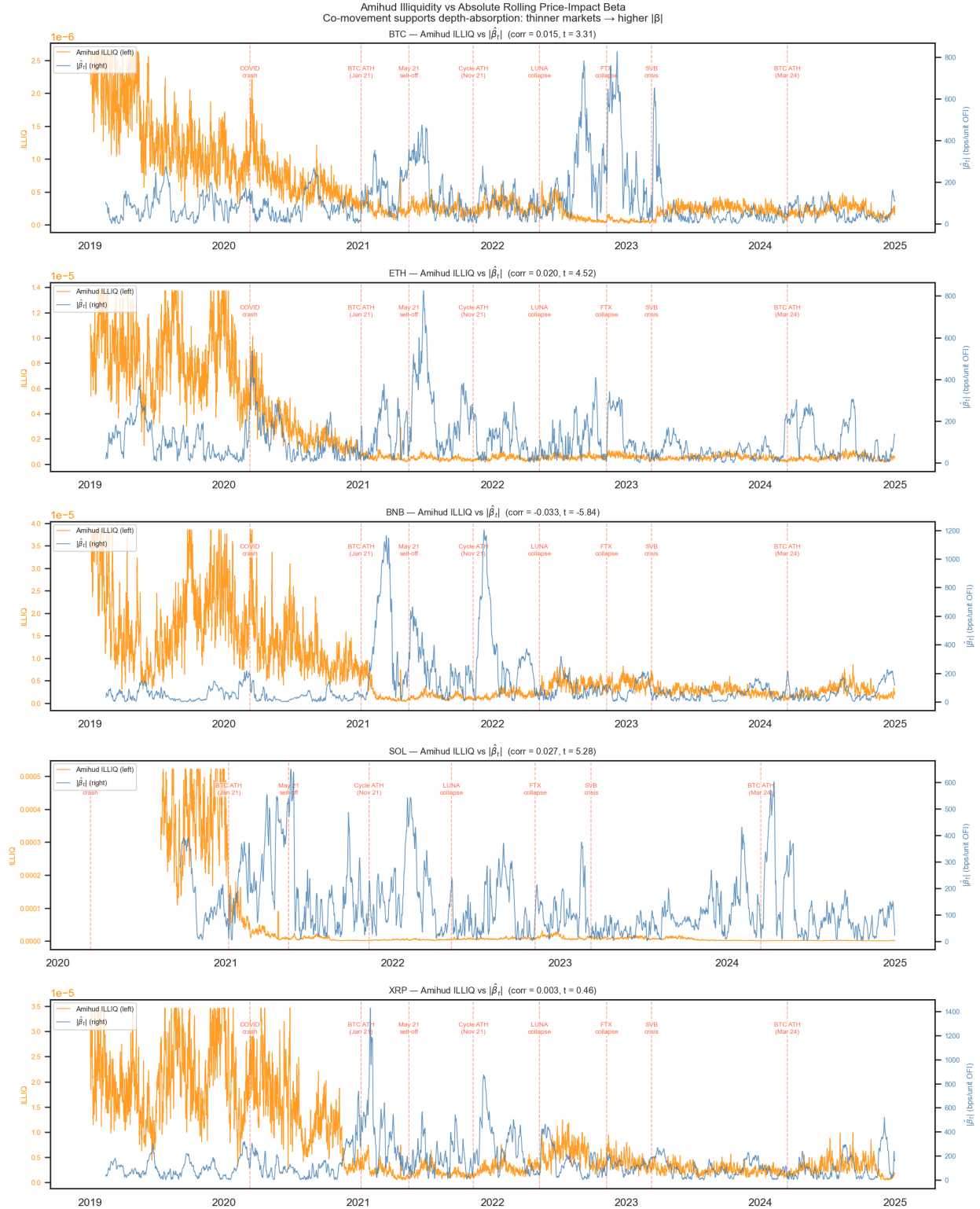


Figure 17: Smoothed Amihud illiquidity ratio $ILLIQ_{i,t}$ (orange, left axis) and smoothed absolute rolling price-impact coefficient $|\hat{\beta}_t|$ (blue, right axis) for each of the five assets over the full sample 2019–2024. Both series are smoothed with a 10-bar moving average. The co-movement between the two series across all assets is consistent with the depth-absorption mechanism: thinner markets exhibit larger price-impact coefficients, amplifying the transmission of multiplicative OFI dynamics into return variance. Vertical dashed lines mark the eight annotated market events.

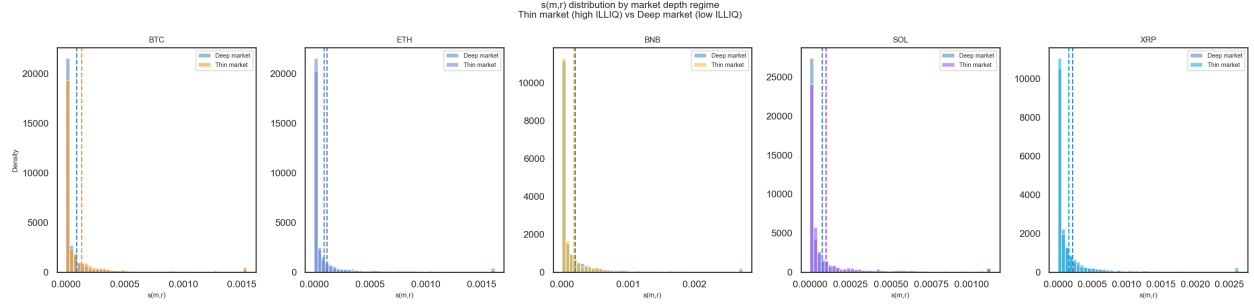


Figure 18: Distribution of the OFI-induced multiplicative return-variance share $\hat{s}_t^{(m,r)}$ conditional on market depth for each asset. Thin-market bars (Amihud ILLIQ_{i,t} above the 70th percentile, coloured) exhibit a higher mean $\hat{s}_t^{(m,r)}$ than deep-market bars (blue), consistent with the depth-absorption mechanism: concentrated order flow is more likely to transmit into return variance when market depth is insufficient to absorb it. Dashed vertical lines mark the within-regime means.

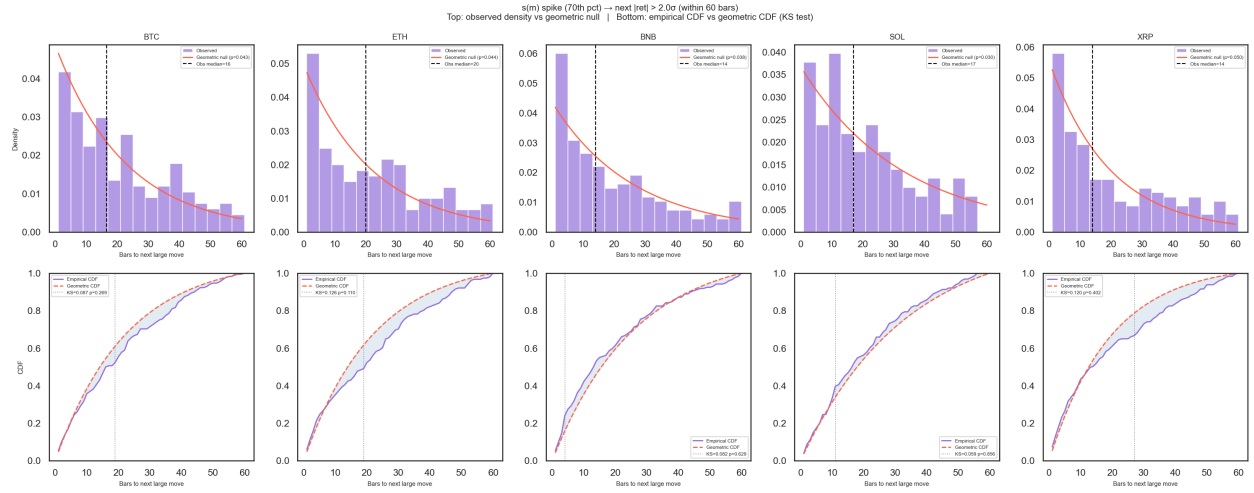


Figure 19: Forward-distance distributions from $\hat{s}_t^{(m)}$ spike onsets (70th percentile threshold) to the next large price move ($|r_t| > 2\hat{\sigma}_t$, within $H = 60$ bars). Top row: observed density (purple bars) against the fitted geometric null PMF (red curve). Bottom row: empirical CDF (purple) against the geometric null CDF (red dashed), with the maximum KS gap shaded in blue and the KS statistic and permutation p -value annotated. The geometric null is fitted to the unconditional large-move arrival rate $\hat{p}_{\text{geom}}$ for each asset.